Design Document

for

Campus Connect

Version 1.0

Prepared by

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Course: CS253

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Date: 04-02-2026

CONTENTS ..... **II**

REVISIONS ..... **II**

1 CONTEXT DESIGN **1**

1.1 CONTEXT MODEL 1
1.2 HUMAN INTERFACE DESIGN 2

2 ARCHITECTURE DESIGN **9**

3 OBJECT ORIENTED DESIGN **11**

3.1 USE CASE DIAGRAMS 11
3.2 CLASS DIAGRAMS 15
3.3 SEQUENCE DIAGRAMS 16
3.4 STATE DIAGRAMS 25

4 PROJECT PLAN **33**

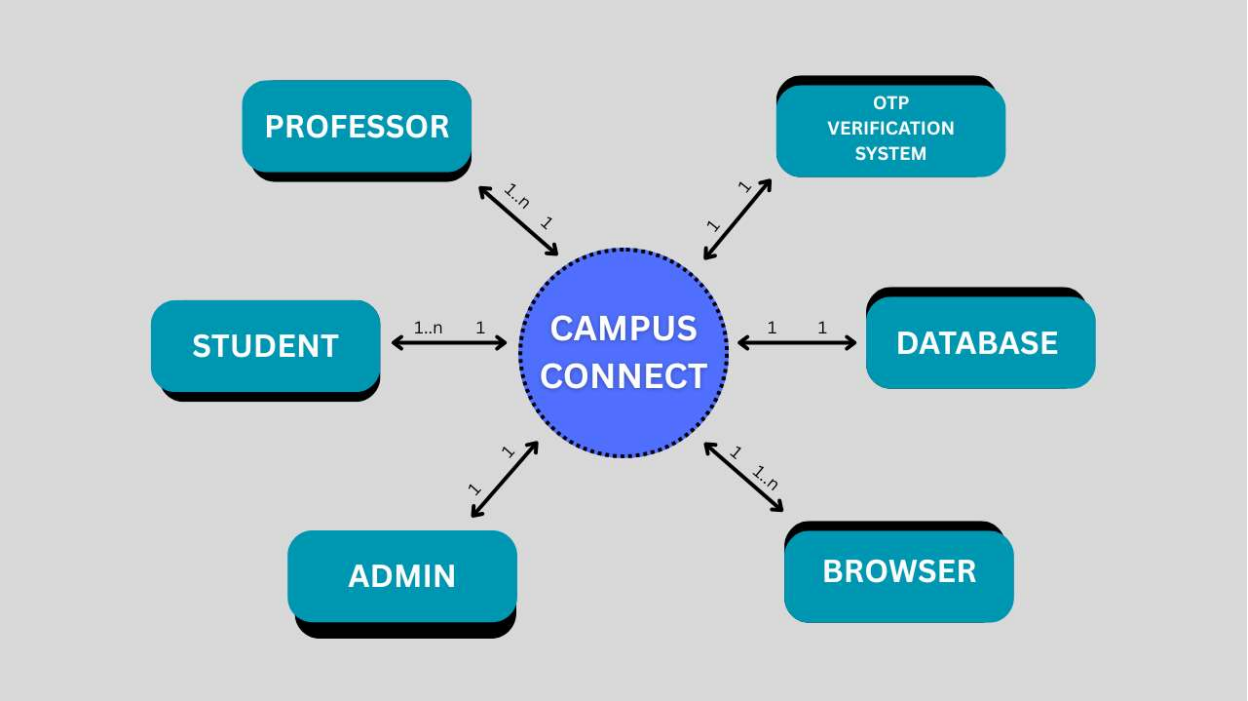
APPENDIX A - GROUP LOG **34**

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
v1.0	Prabhakar Raj Navdeep Singh Raj Vatsal Kumar Dishank Kumawat Mrityunjay Kumar Rupam Das Prateek Raj Karan Kumar Sumit Vishwakarma	The first version of the design document	04/02/26

1 Context Design

1.1 Context Model



• **Student** : Students interact with the Campus Connect system to register/login, view enrolled courses, submit anonymous feedback for each class session, post comments and replies, and apply for project opportunities while tracking their application status.

• **Professor** : Professors use the system to register/login, create and manage courses, automatically generate class sessions, upload student enrollment lists, publish project opportunities, review and filter applications, accept/reject applications, and view course feedback analytics.

• **Admin** : The admin manages professor authorization by adding or removing professor email IDs in the system. Only professors approved by the admin are allowed to register and access professor functionalities.

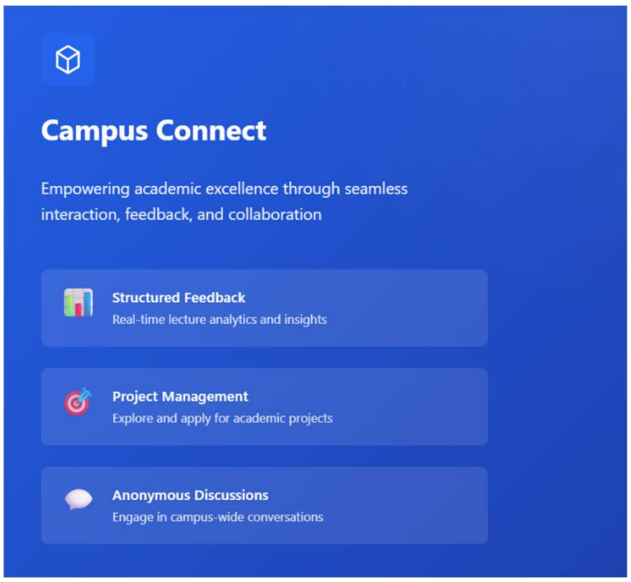
• **OTP Verification System** : The OTP verification system is responsible for verifying user email addresses during sign-up and password recovery. It sends OTP codes and validates them to ensure secure authentication.

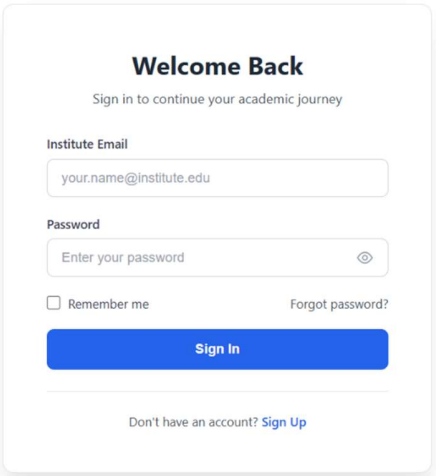
• **Database** : The database serves as the central storage for all system data, including user accounts, course details, class sessions, student enrollment records, anonymous feedback submissions, discussion messages, project listings, student applications, and application status updates.

• **Browser** : The browser acts as the interface through which students, professors, and admins access the Campus Connect system, enabling interaction with all system features through a web-based platform.

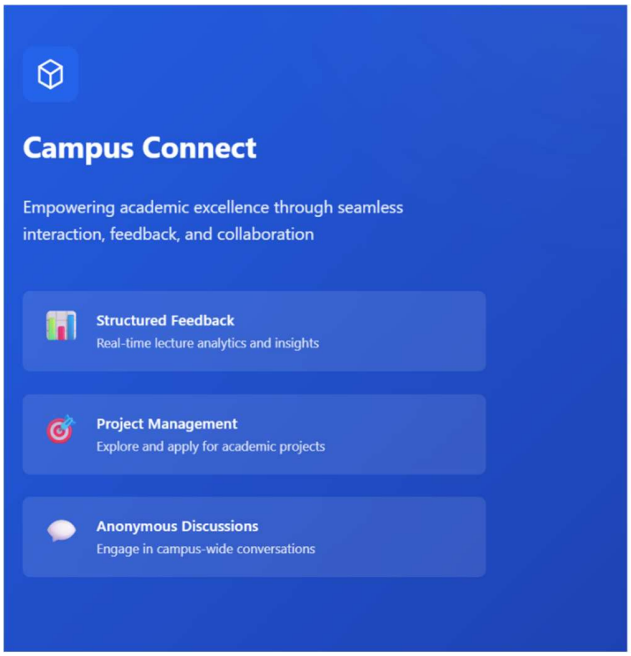
1.2 Human Interface Design

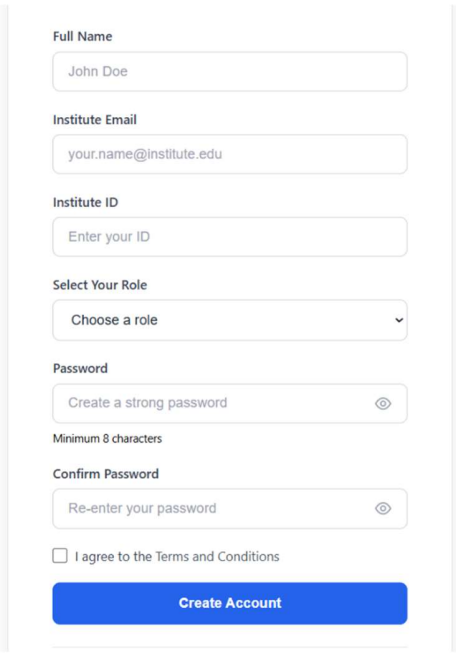
1.2.1 Login :



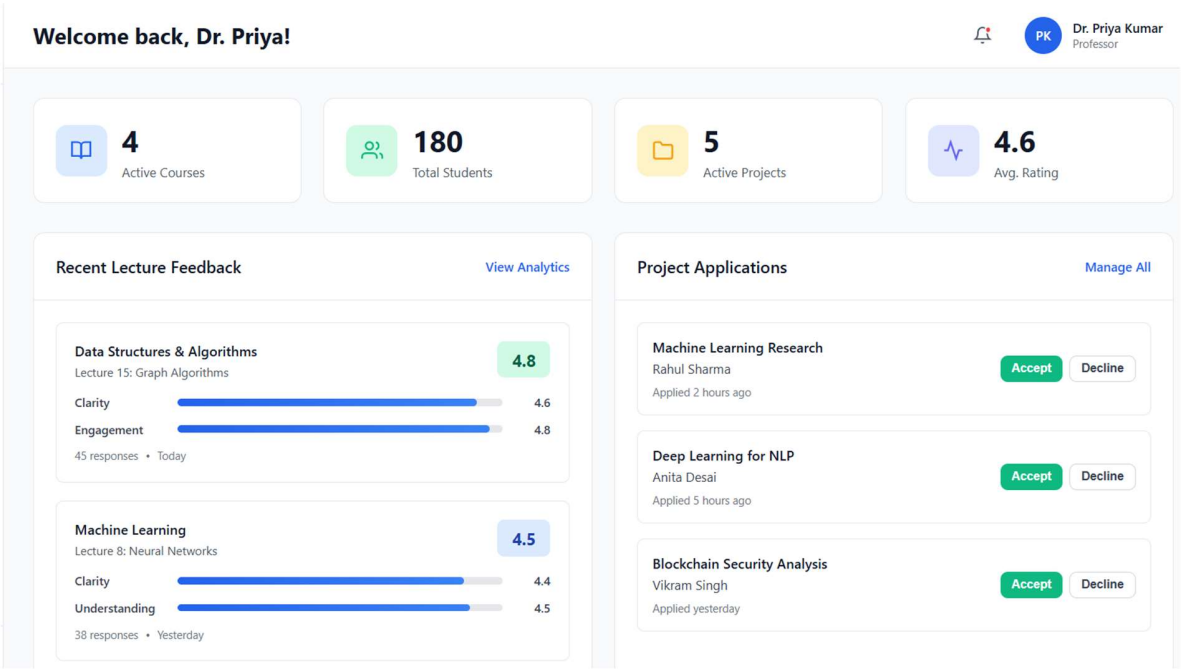


1.2.2 Sign Up:

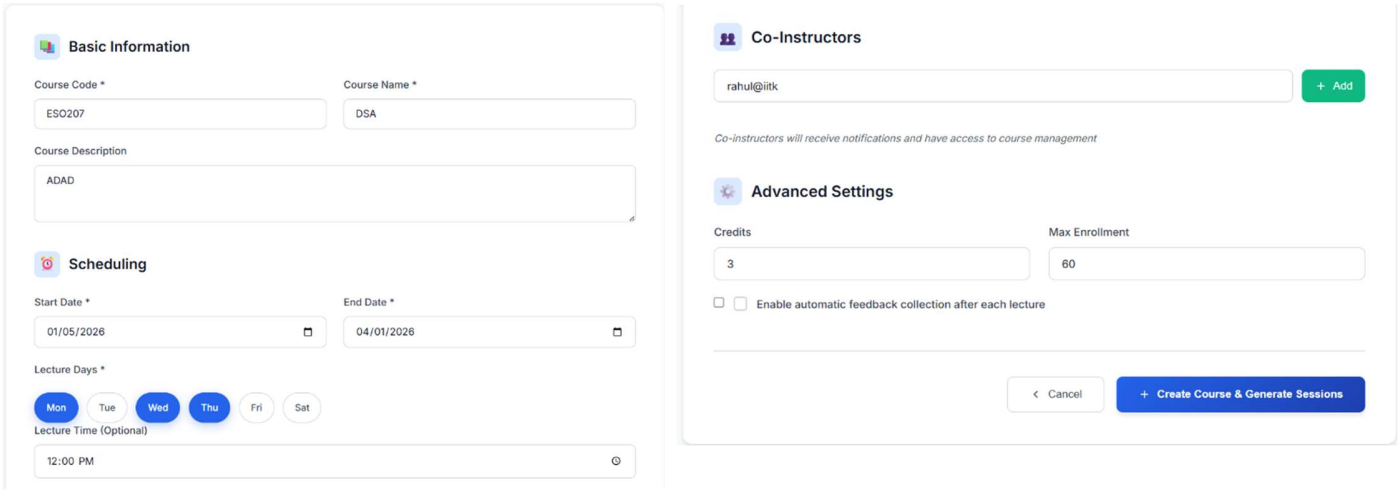




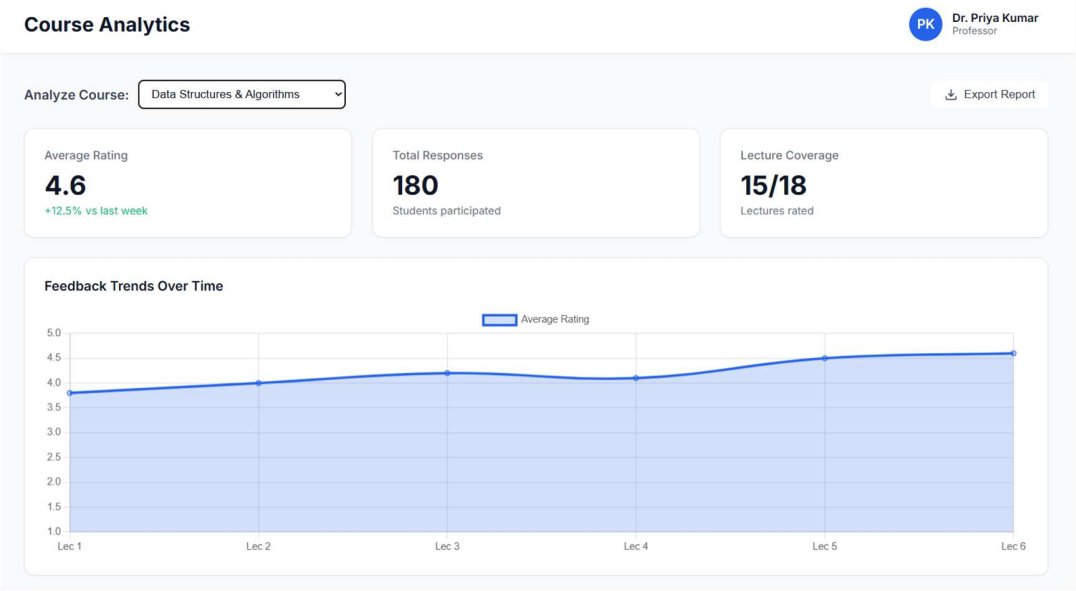
1.2.3 Professor Dashboard

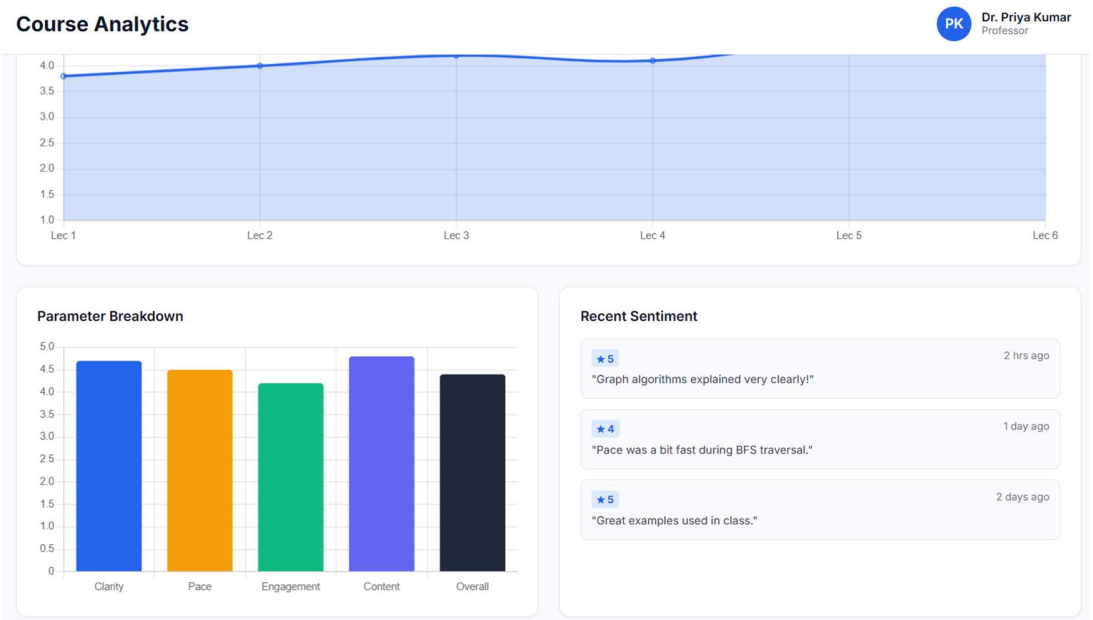


1.2.4 Course Creation

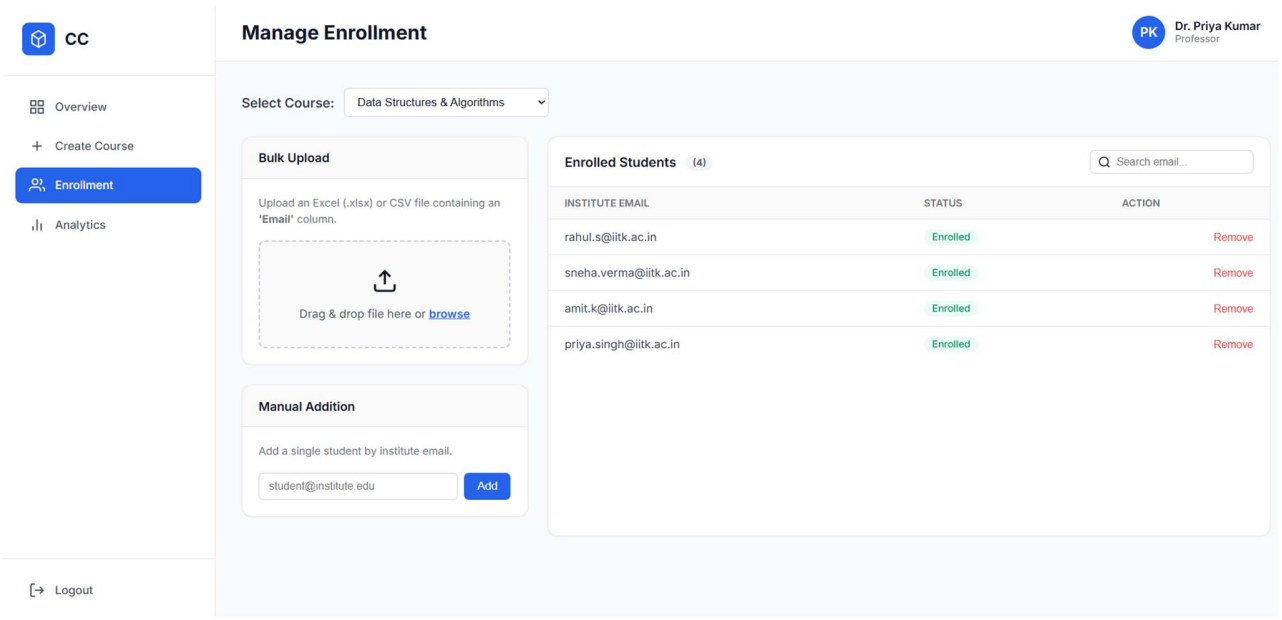


1.2.5 Course-wise Feedback Analytics

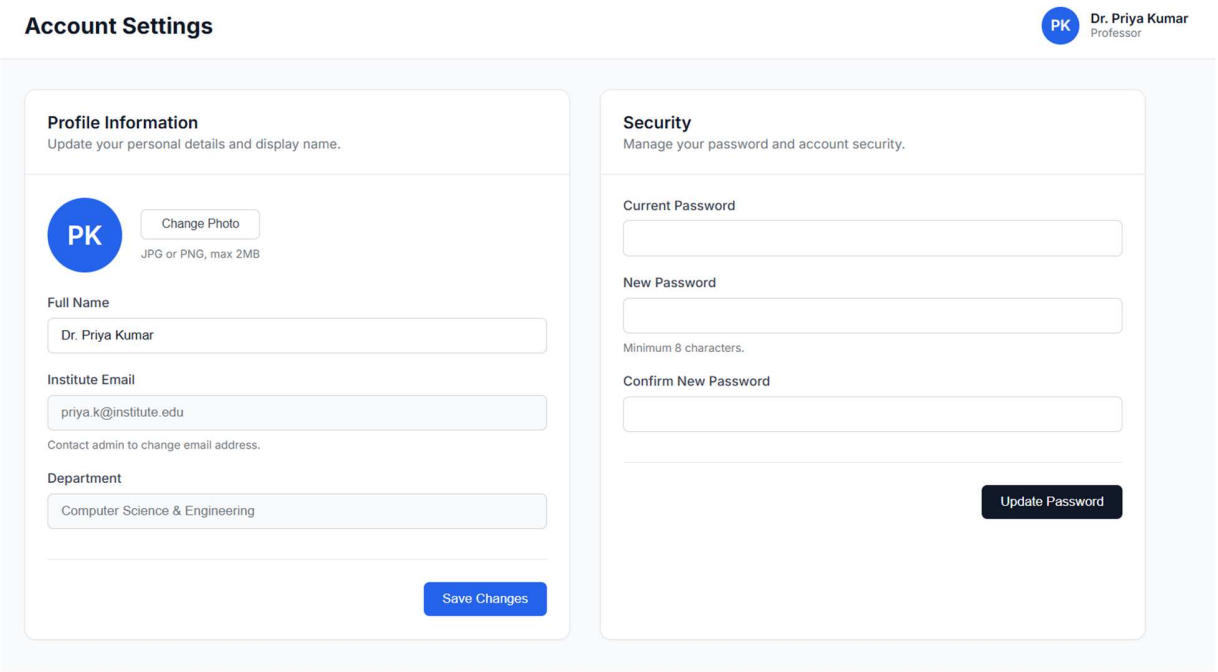




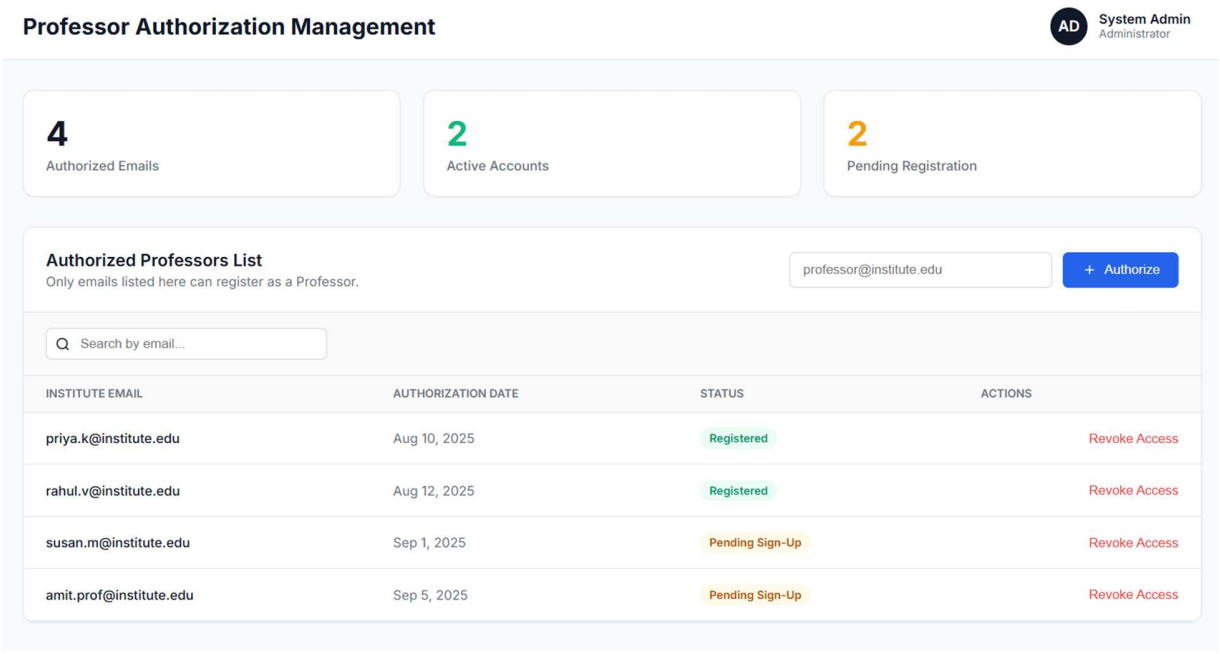
1.2.6 Student Enrollment



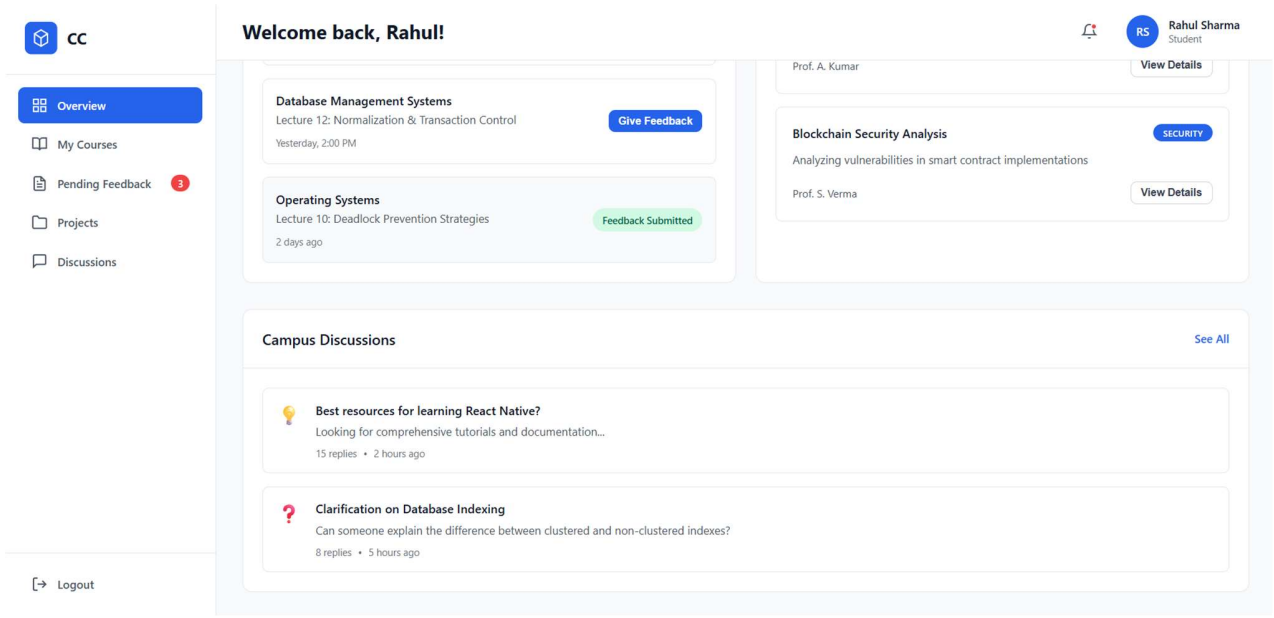
1.2.7 Profile Settings



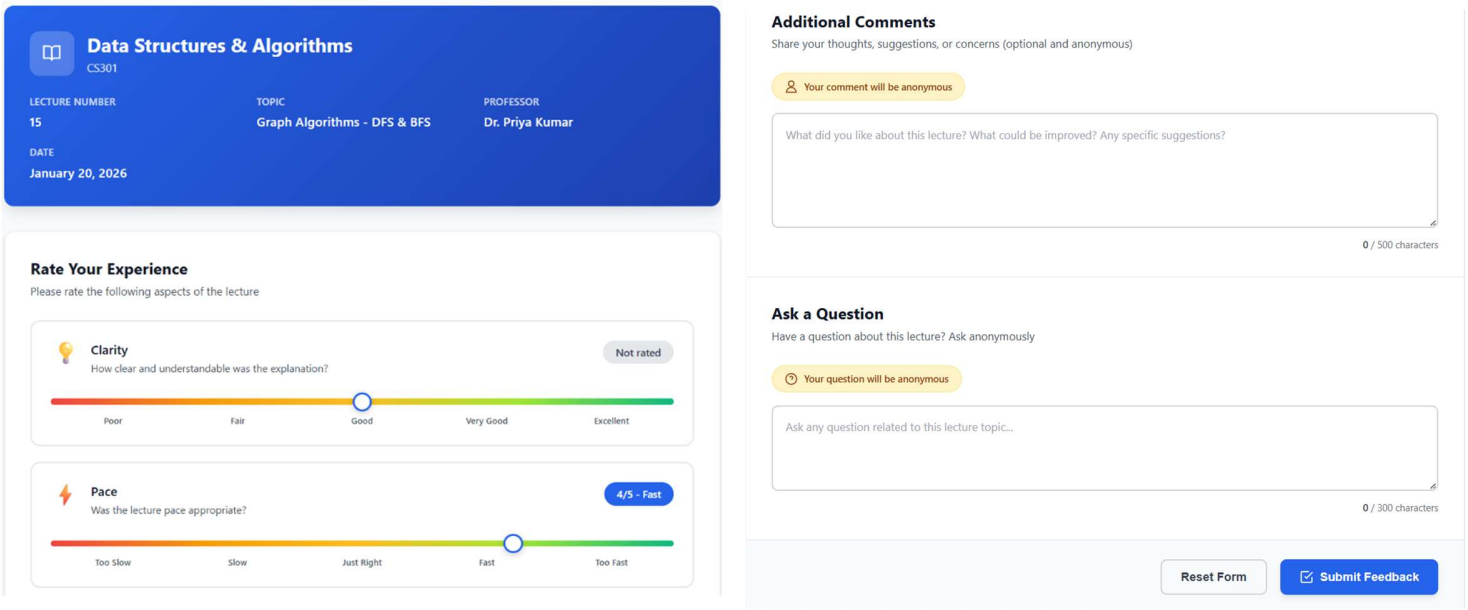
1.2.8 System Admin Dashboard



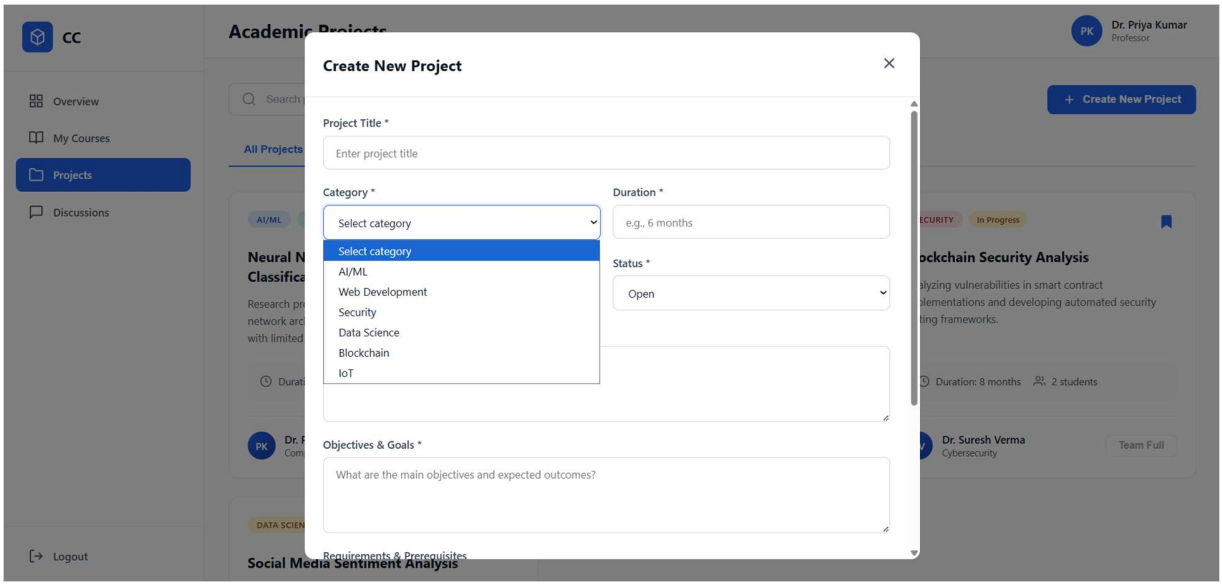
1.2.9 Student Dashboard



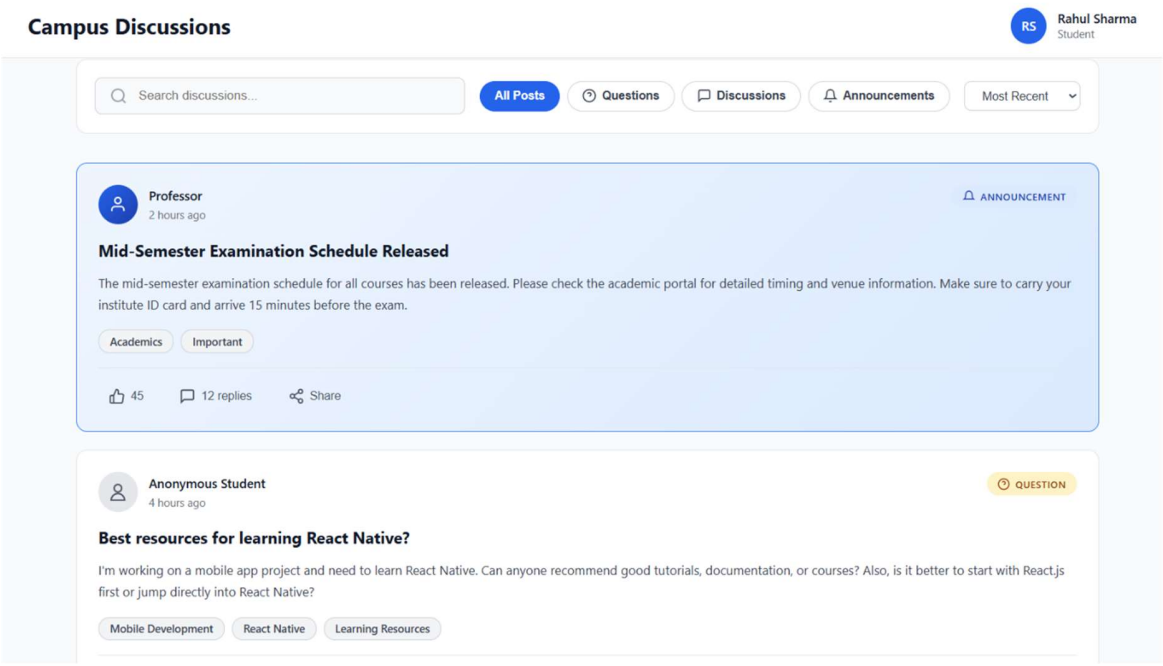
1.2.10 Lecture Feedback



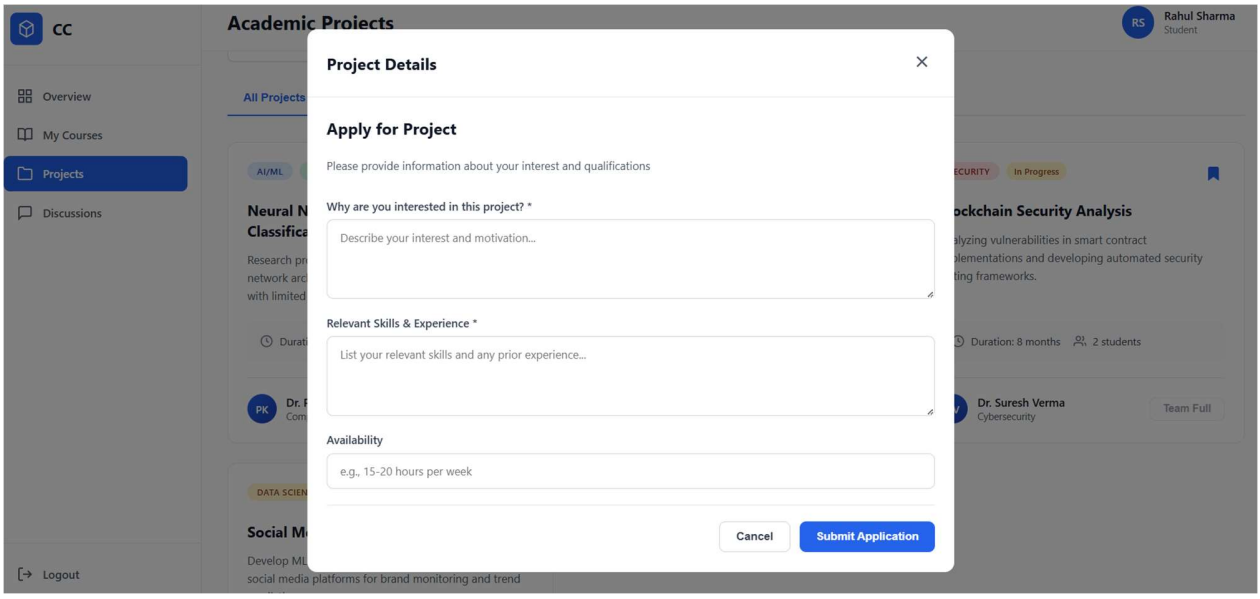
1.2.11 Project Creation



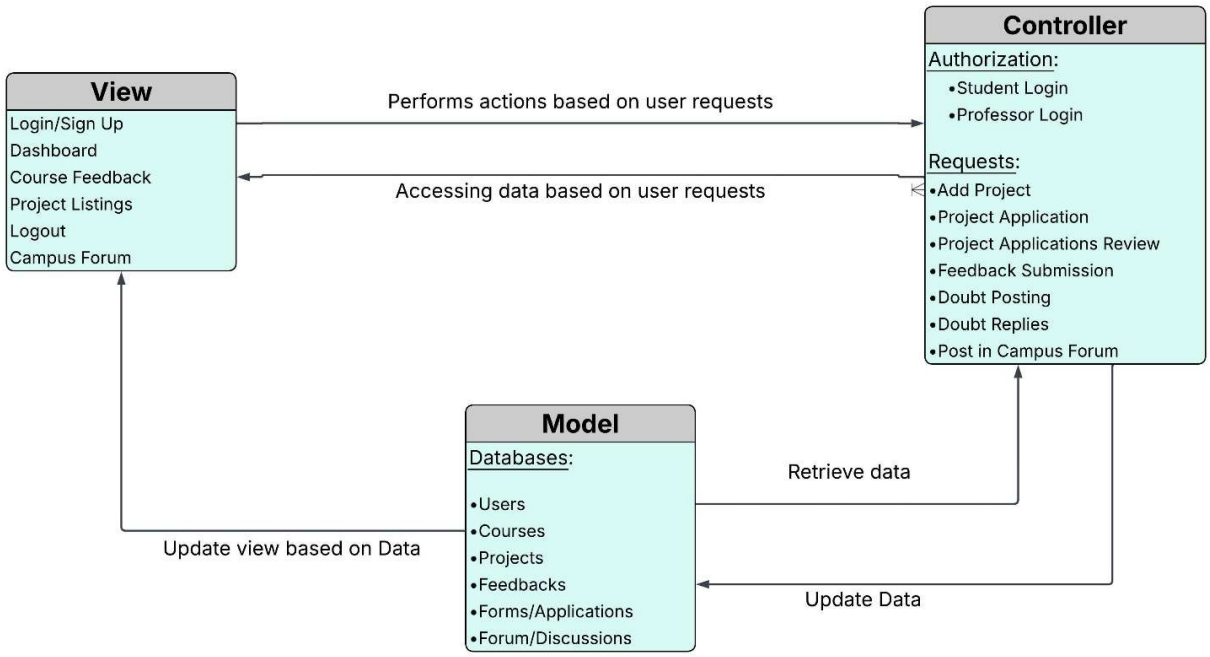
1.2.12 Anonymous Discussions



1.2.13 Project Application



2 Architecture Design



MODEL VIEW CONTROLLER ARCHITECTURE

The Campus Connect utilizes a Model-View-Controller architecture to carry out operations related to academic project management, course feedback collection, and campus-wide discussions. It separates the platform into three logical components: the model, the view, and the controller. Each component is built to handle specific aspects of the interaction between students and professors.

MODEL

The Model is responsible for managing the central database. It handles the storage and retrieval of critical academic data, including **User profiles (Students/Professors), Course details, Project listings and Feedback forms.**

It receives instructions from the Controller regarding new project applications or forum posts and updates the database accordingly. It also retrieves requested data, such as a list of available projects, to be displayed by the View.

VIEW

The View component defines the user interface for the campus portal. It handles the presentation layer, providing the visual dashboard where users can **log in, browse project listings, submit course and classes feedback, and participate in the campus forum.**

The View acts as the front-end layer that users interact with directly. It captures user inputs (like a "Project Application") and sends them to the Controller, while dynamically displaying data retrieved from the Model.

CONTROLLER

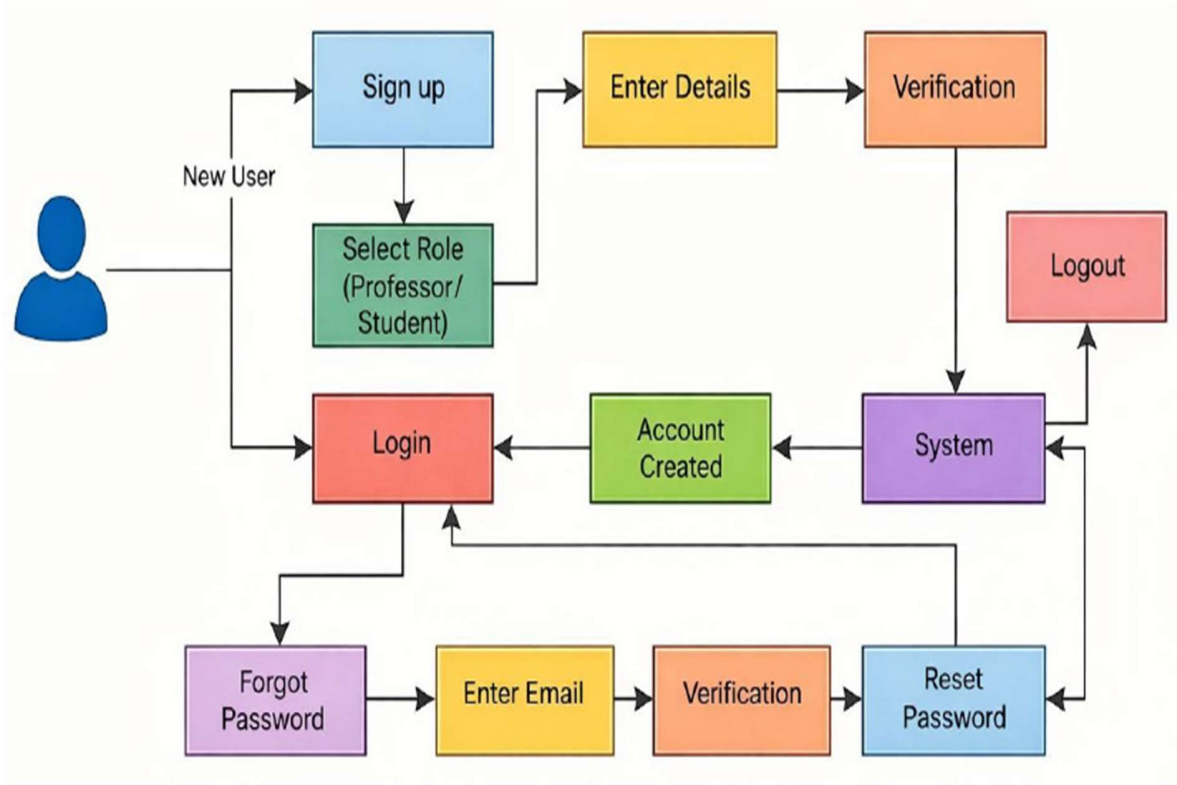
The Controller acts as the logical intermediary between the Model and View. It manages the authorization logic, distinguishing between **Student and Professor logins** to grant appropriate access.

It processes specific user requests such as **"Add Project," "Review Application," or "Post Doubt."** The Controller determines the flow of data, ensuring that a student's application is correctly routed to the Model for storage and that the status is updated back to the View.

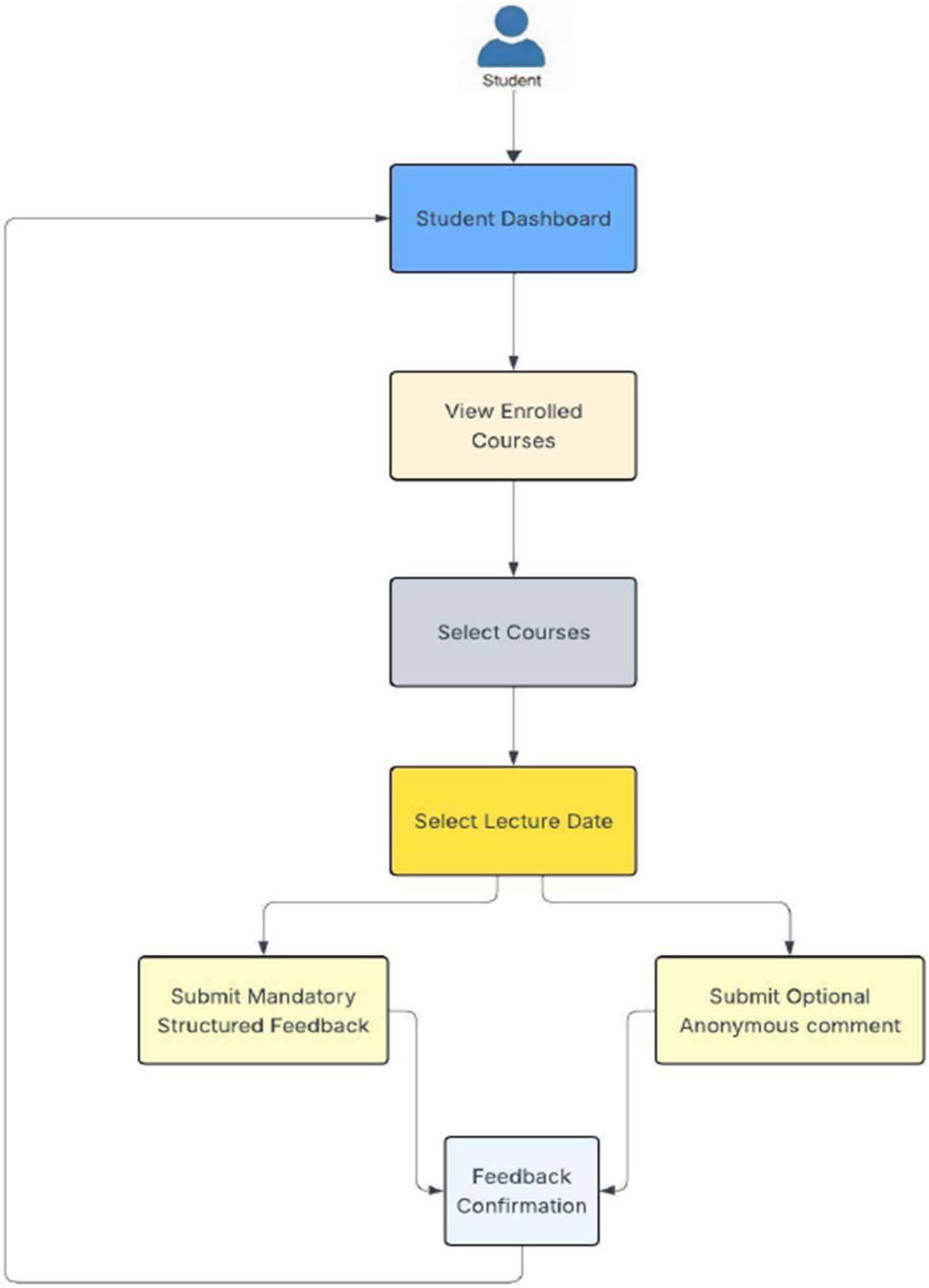
3 Object Oriented Design

3.1 Use Case Diagrams

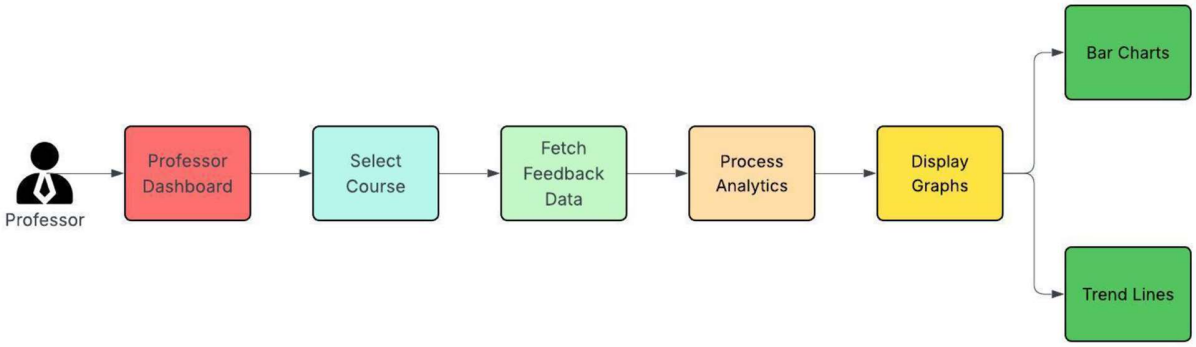
3.1.1 Sign Up/Login



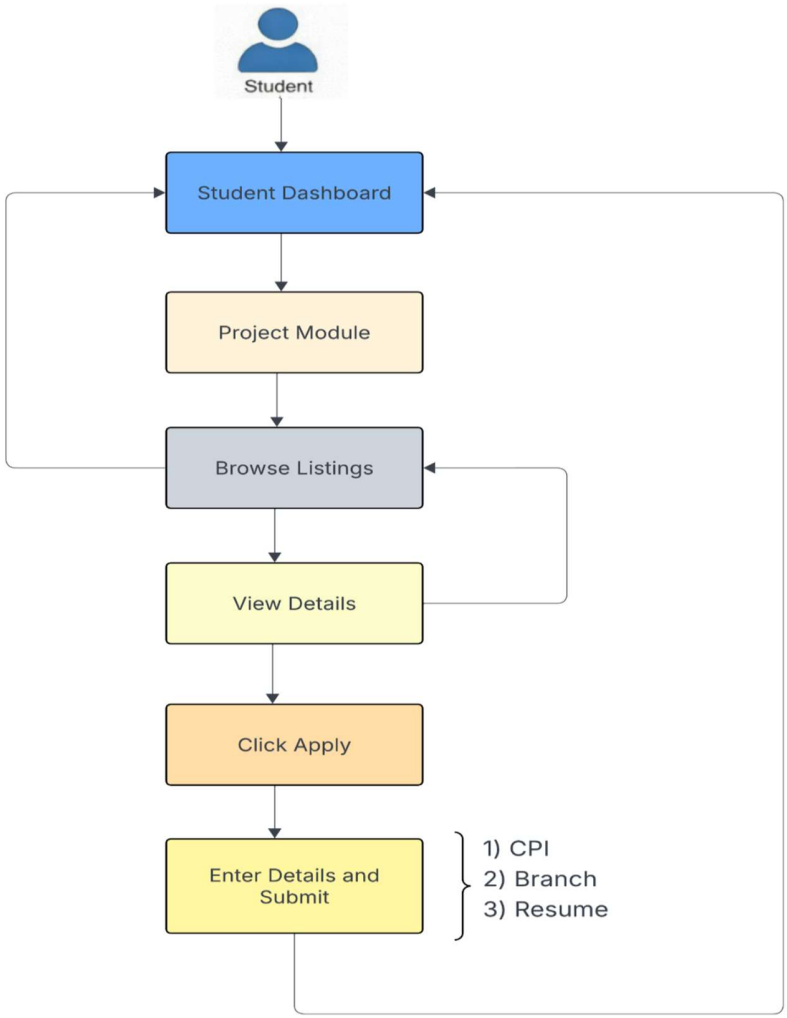
3.1.2 Submit Lecture Feedback



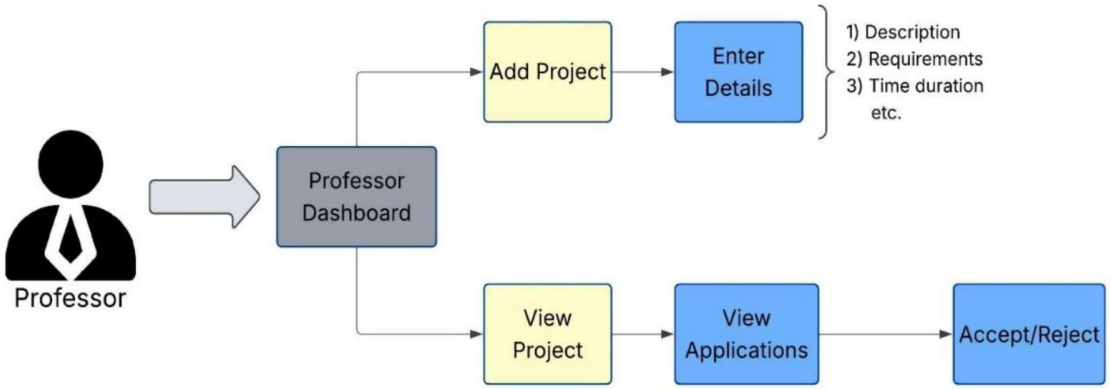
3.1.3 View Course Analytics



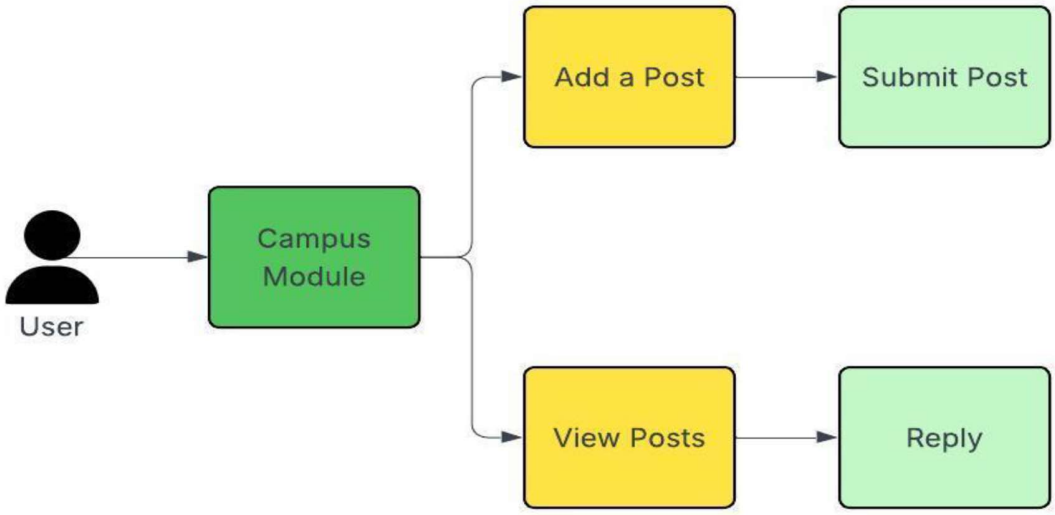
3.1.4 Student Project Module



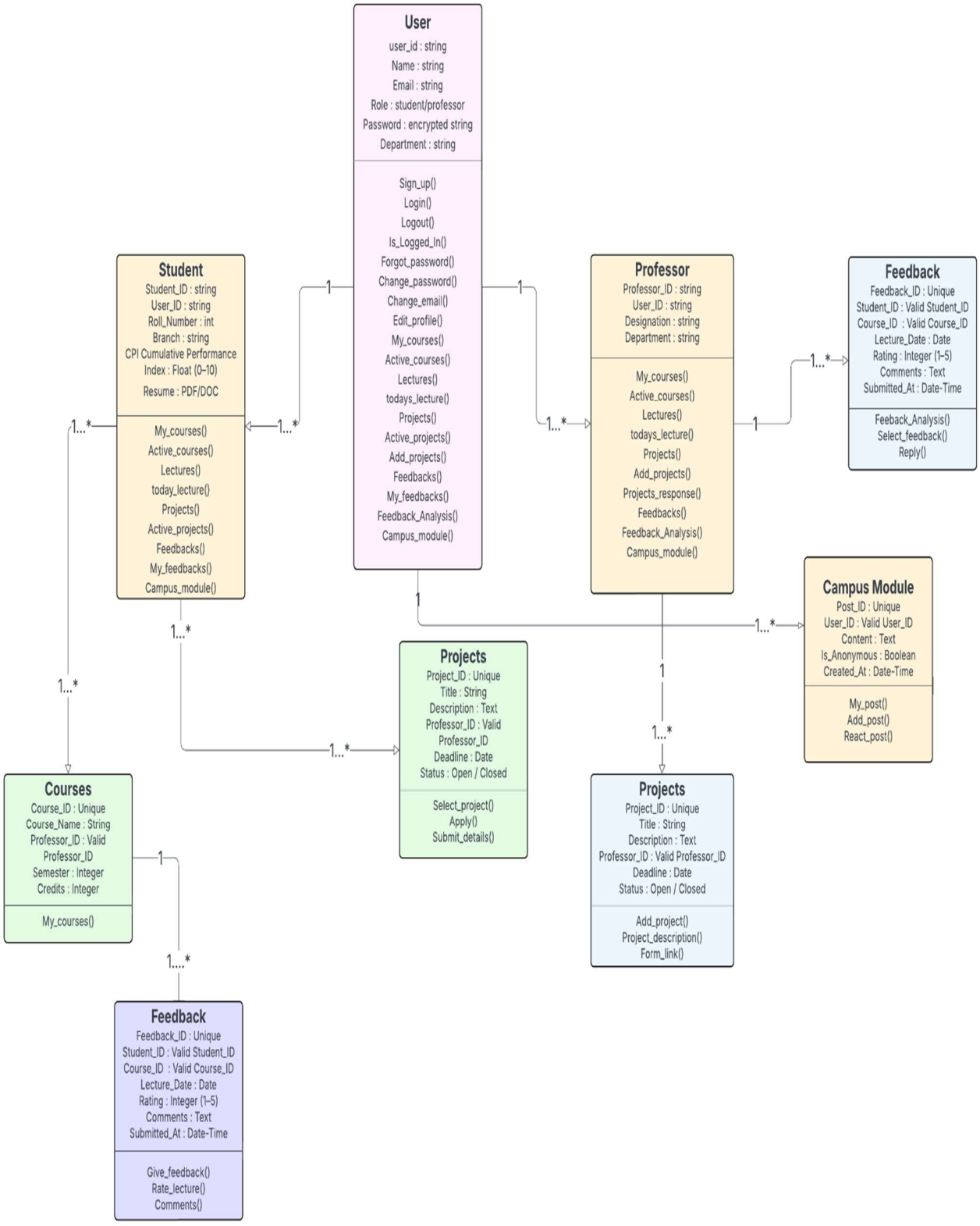
3.1.5 Professor Project Module



3.1.6 Campus Interaction Module



3.2 Class Diagrams

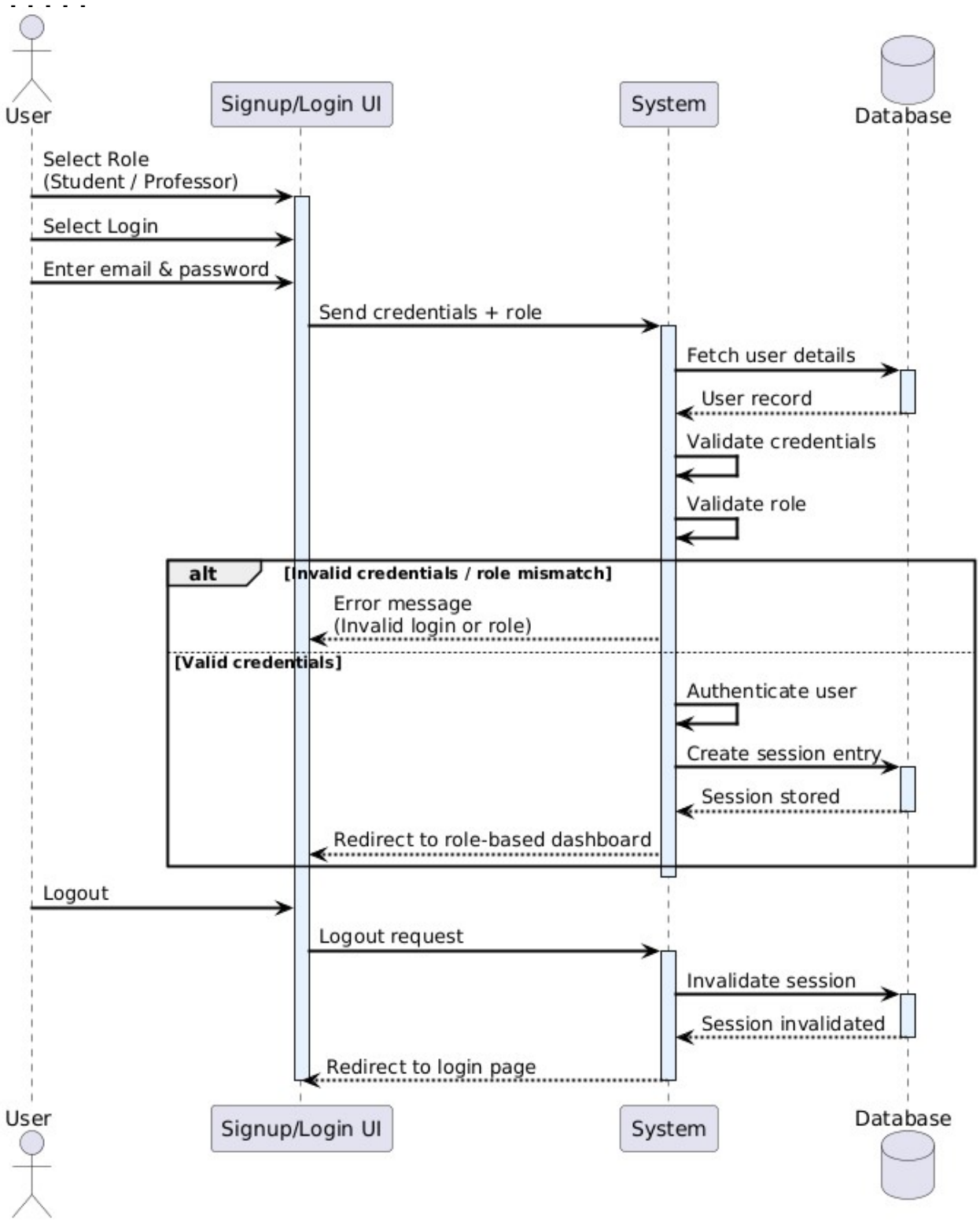


3.3 Sequence Diagrams

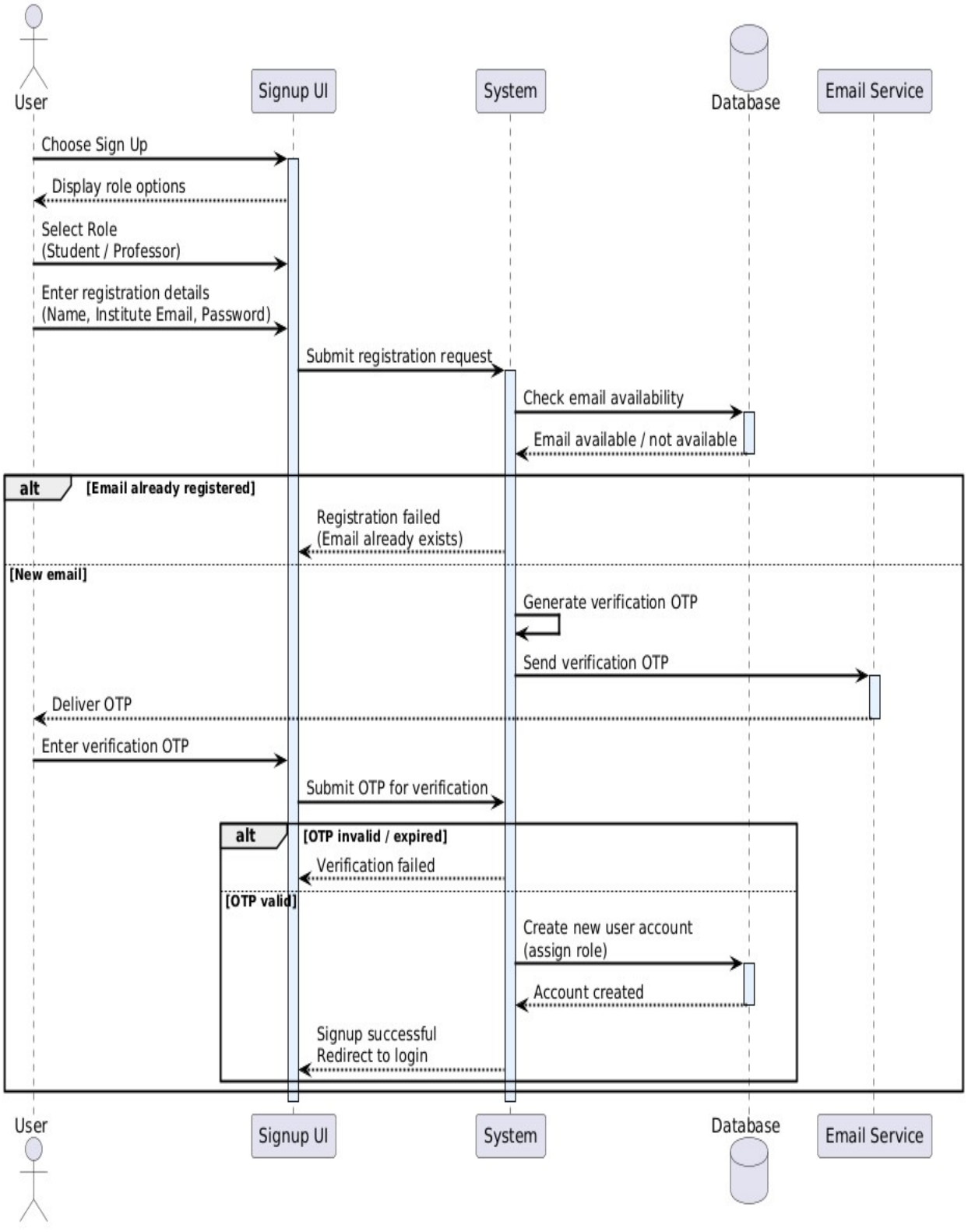
UML Sequence Diagram Symbols - Legend

Symbol	Meaning / Description
Solid Arrow	Synchronous message Sender waits for acknowledgement
Dashed Arrow	Asynchronous message Sender does not wait for reply
«datastore»	Represents a database
abc : xyz	Object *abc* is an instance of class *xyz*
«create»	Creation of an object
✘ (Cross mark)	Destruction of an object
Vertical dashed line	Lifeline symbol (Time progresses downward)
alt	Alternative fragment (Exactly one path executes)
opt	Optional fragment (Executed only if condition is true)
loop	Repetitive interaction fragment

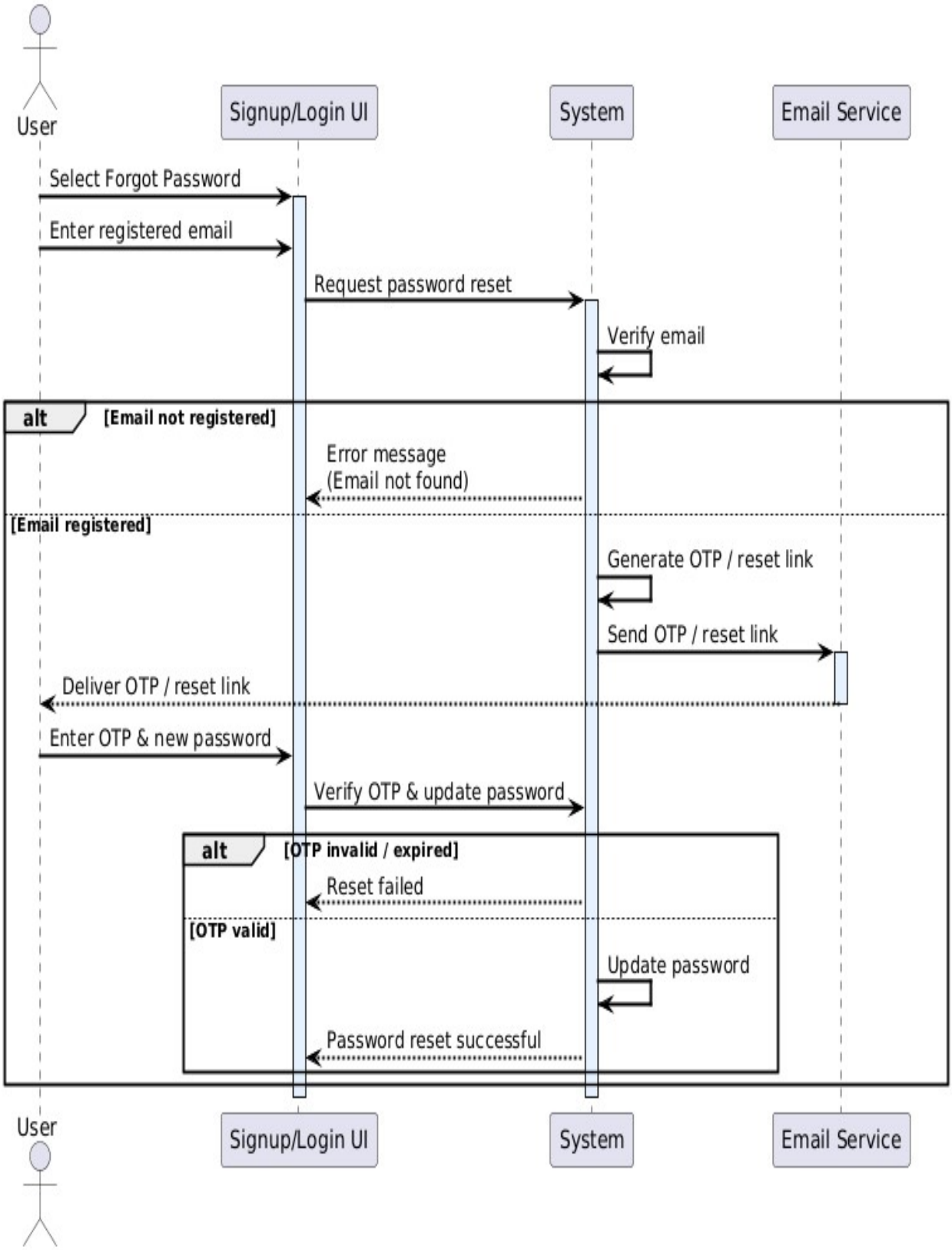
3.3.1 Login


User
Signup/Login UI
System
Database
Select Role
(Student / Professor)
Select Login
Enter email & password
Send credentials + role
Fetch user details
User record
Validate credentials
Validate role
alt
[Invalid credentials / role mismatch]
Error message
(Invalid login or role)
[Valid credentials]
Authenticate user
Create session entry
Session stored
Redirect to role-based dashboard
Logout
Logout request
Invalidate session
Session invalidated
Redirect to login page
User
Signup/Login UI
System
Database

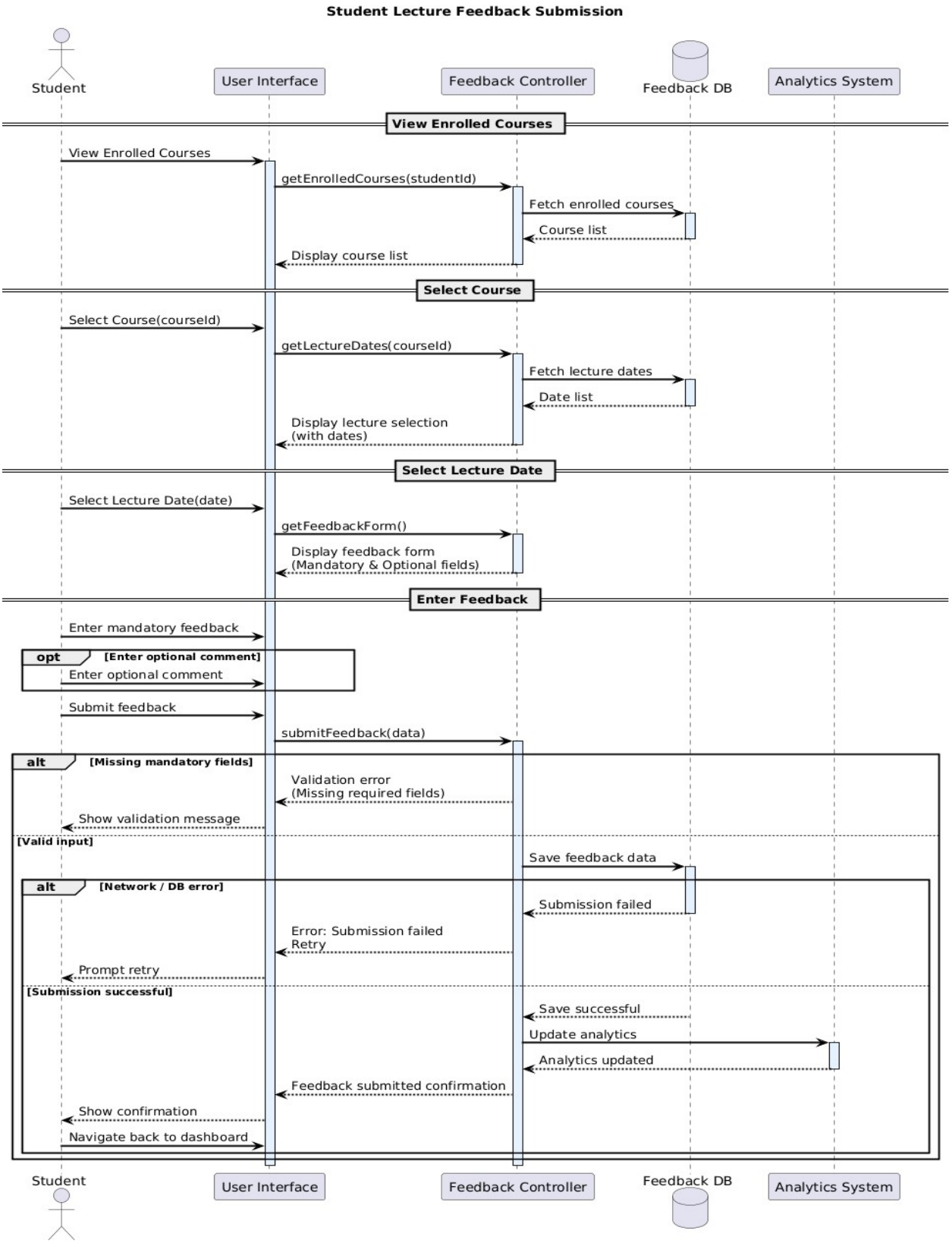
3.3.2 Sign Up


User
Signup UI
System
Database
Email Service
Choose Sign Up
Display role options
Select Role
(Student / Professor)
Enter registration details
(Name, Institute Email, Password)
Submit registration request
Check email availability
Email available / not available
alt
[Email already registered]
Registration failed
(Email already exists)
[New email]
Generate verification OTP
Send verification OTP
Deliver OTP
Enter verification OTP
Submit OTP for verification
alt
[OTP invalid / expired]
Verification failed
[OTP valid]
Create new user account
(assign role)
Account created
Signup successful
Redirect to login
User
Signup UI
System
Database
Email Service

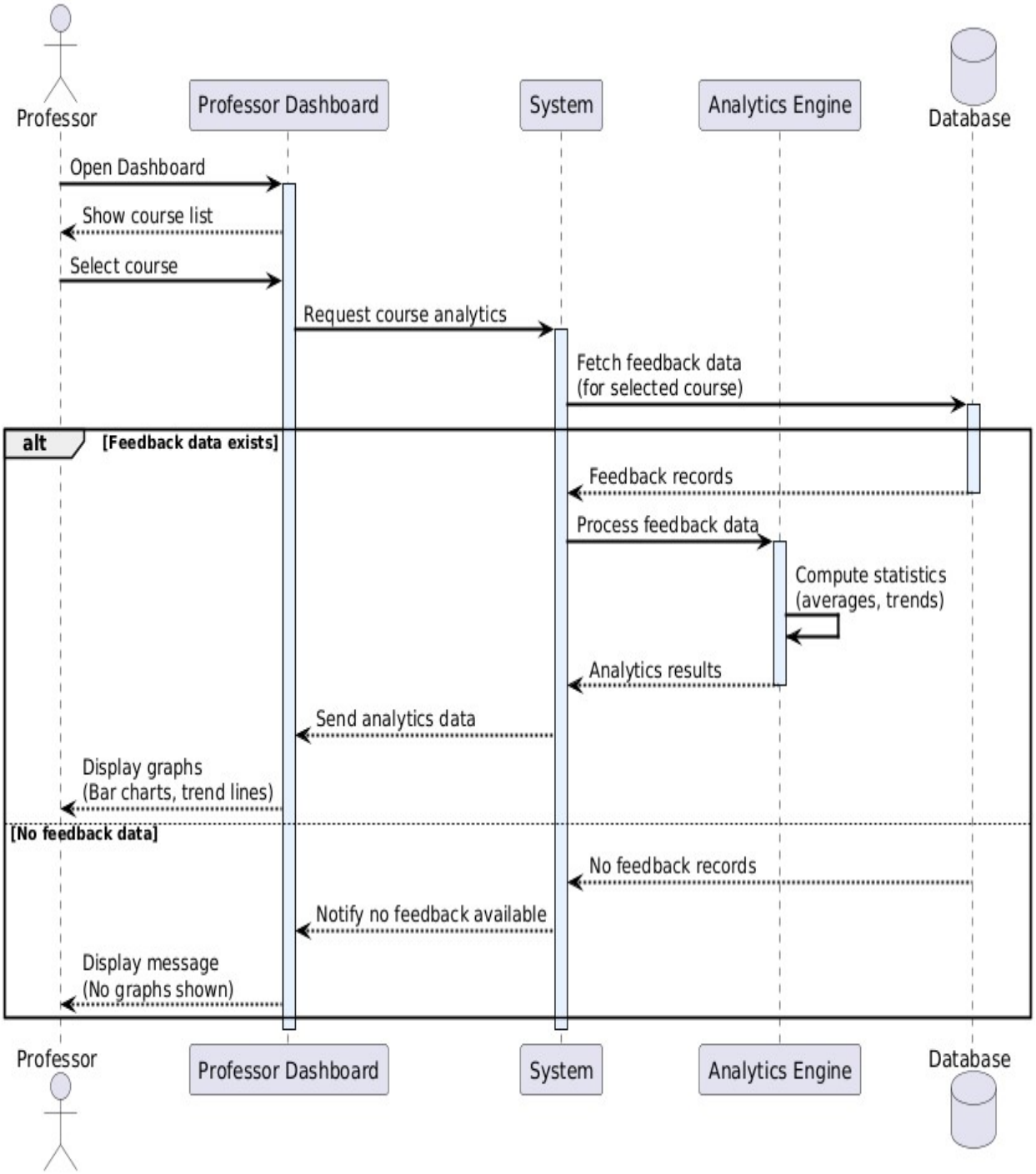
3.3.3 Forgot Password



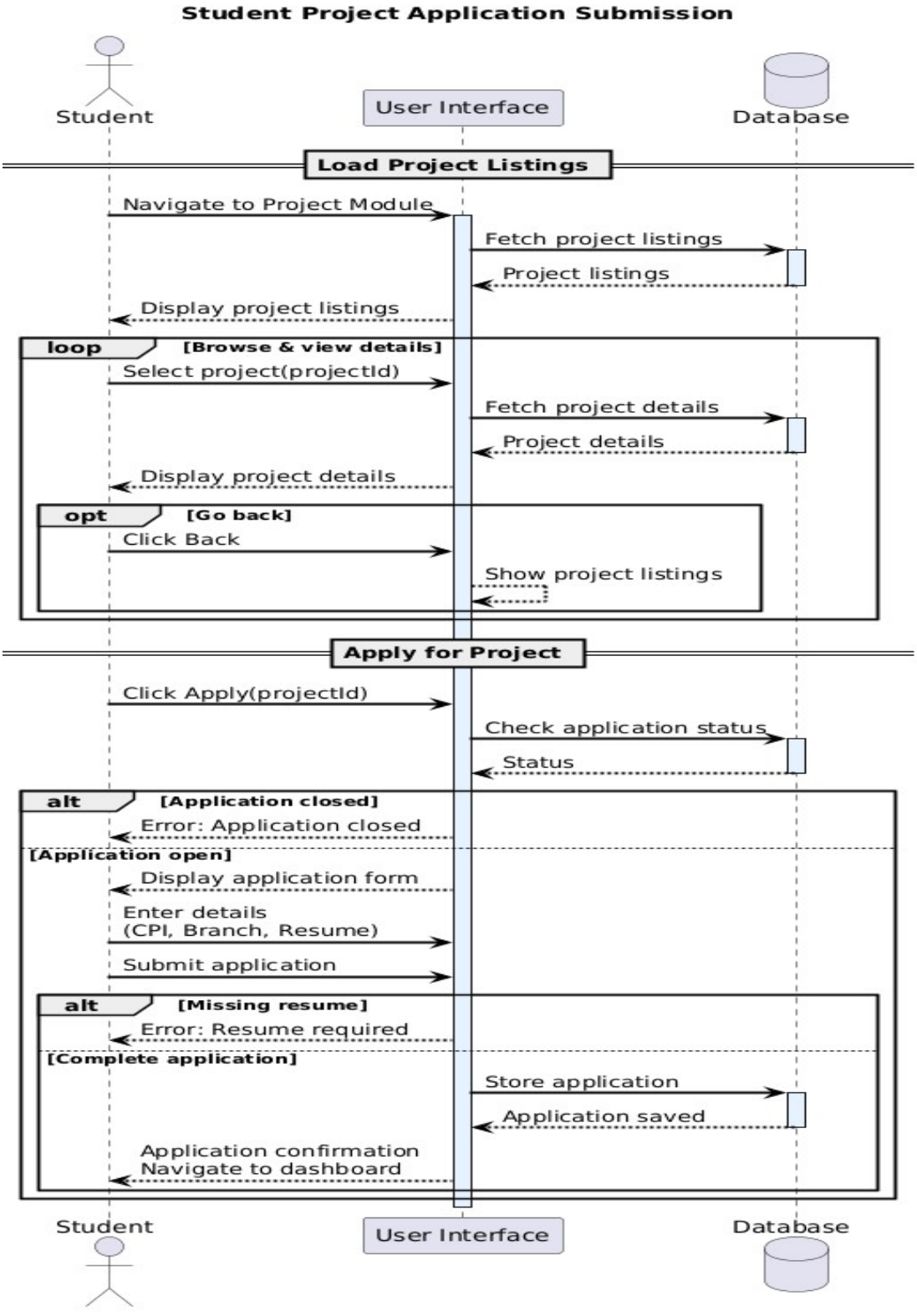
3.3.4 Student Lecture Feedback Submission



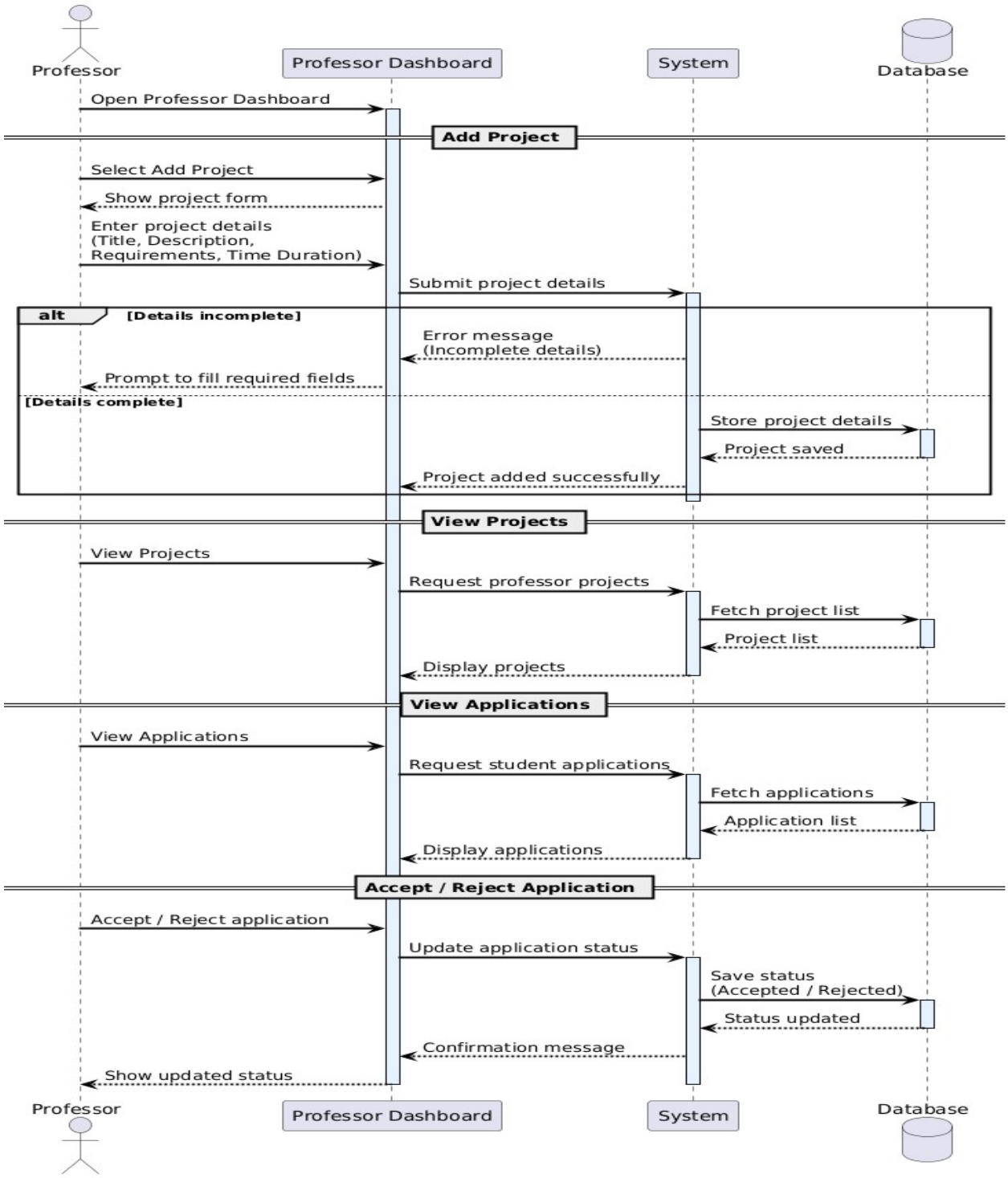
3.3.5 View Course Analytics



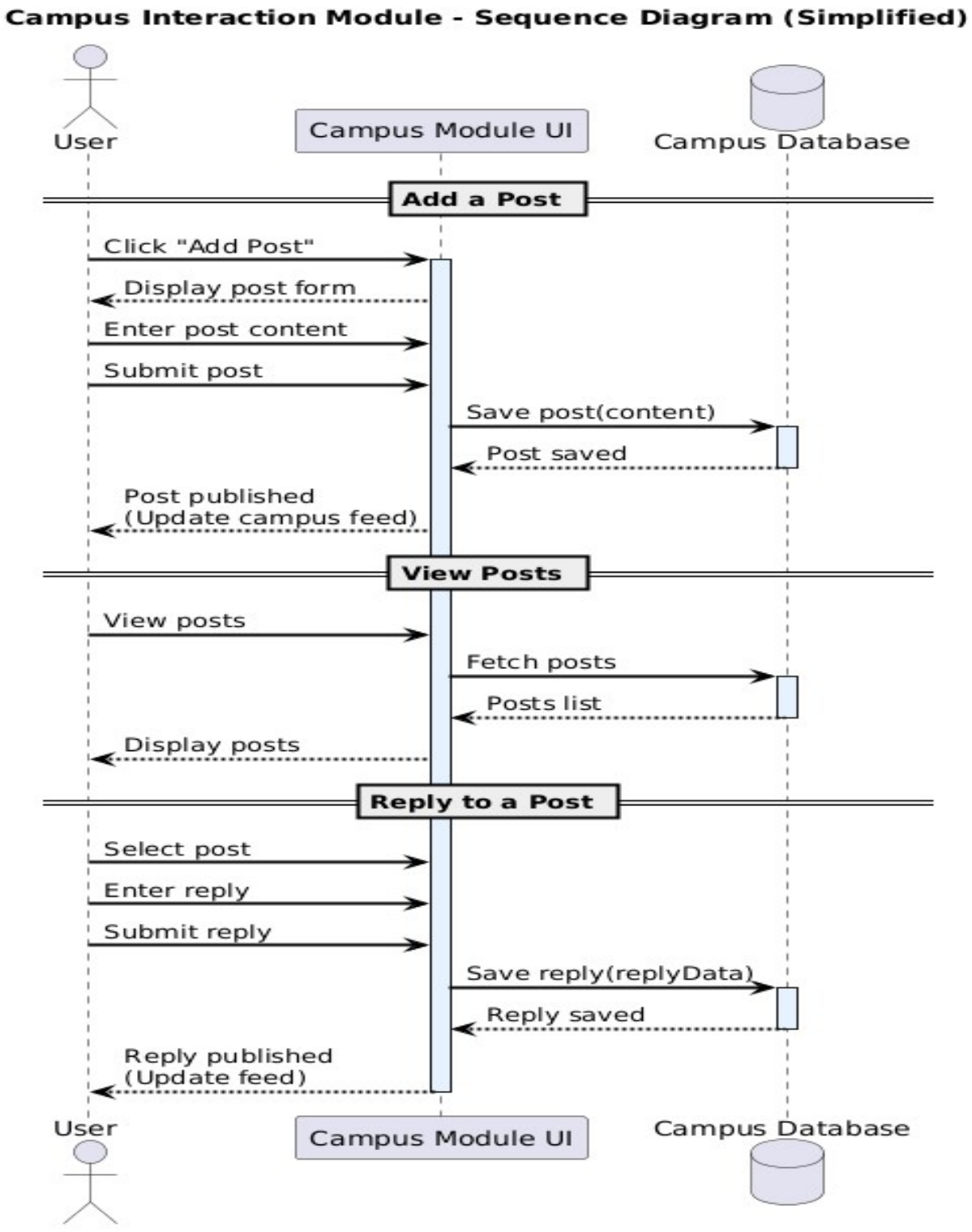
3.3.6 Student Project Module



3.3.7 Professor Project Module

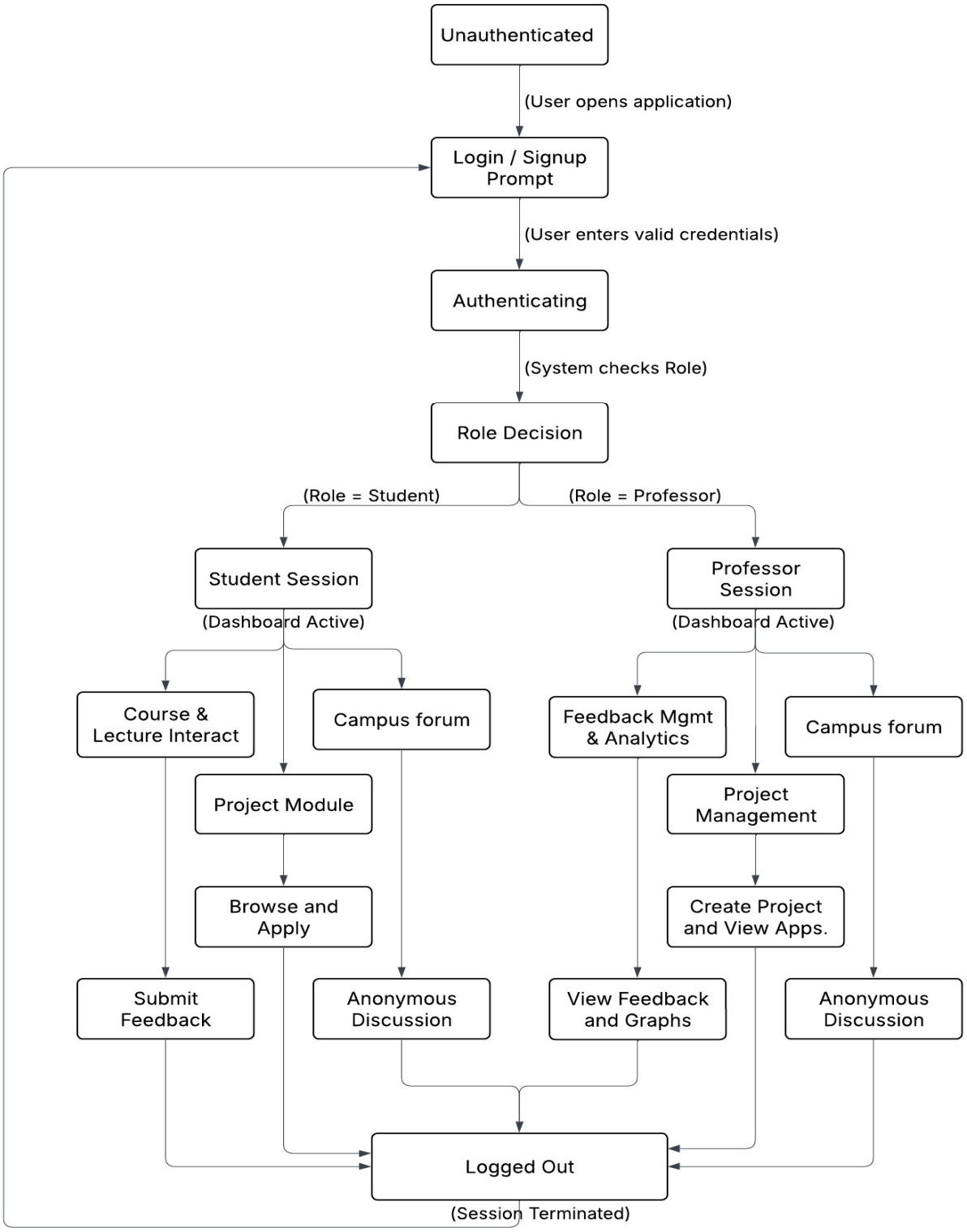


3.3.8 Campus Interaction Module

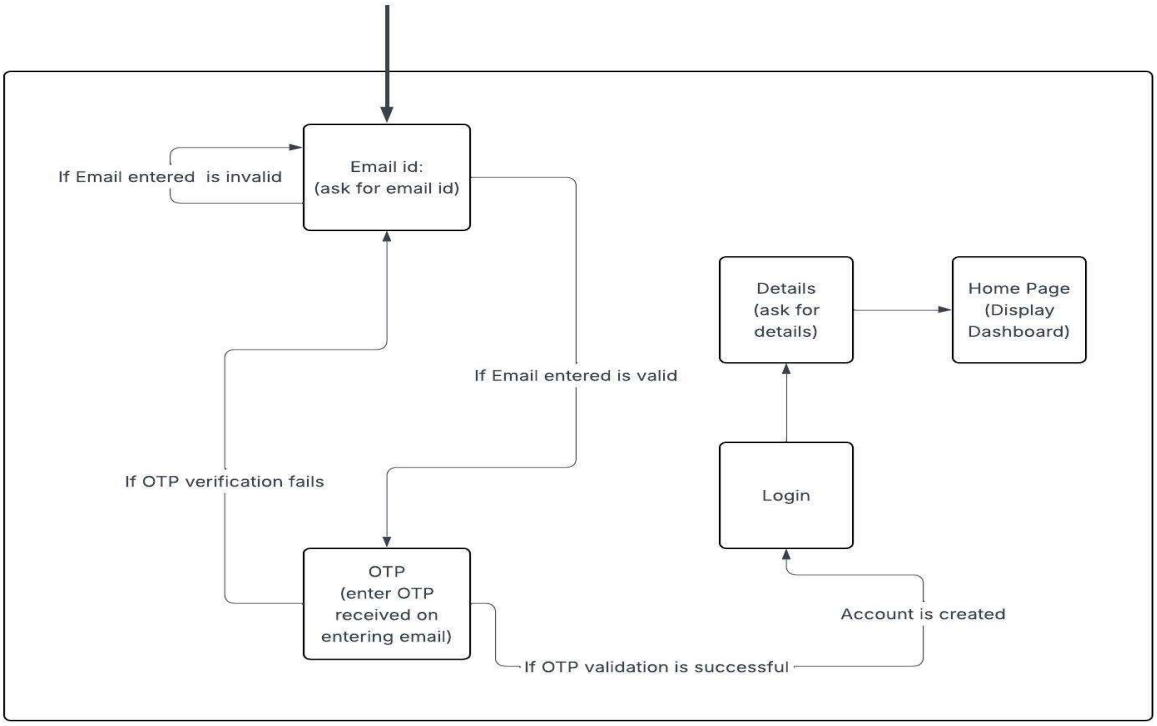


3.4 State Diagrams

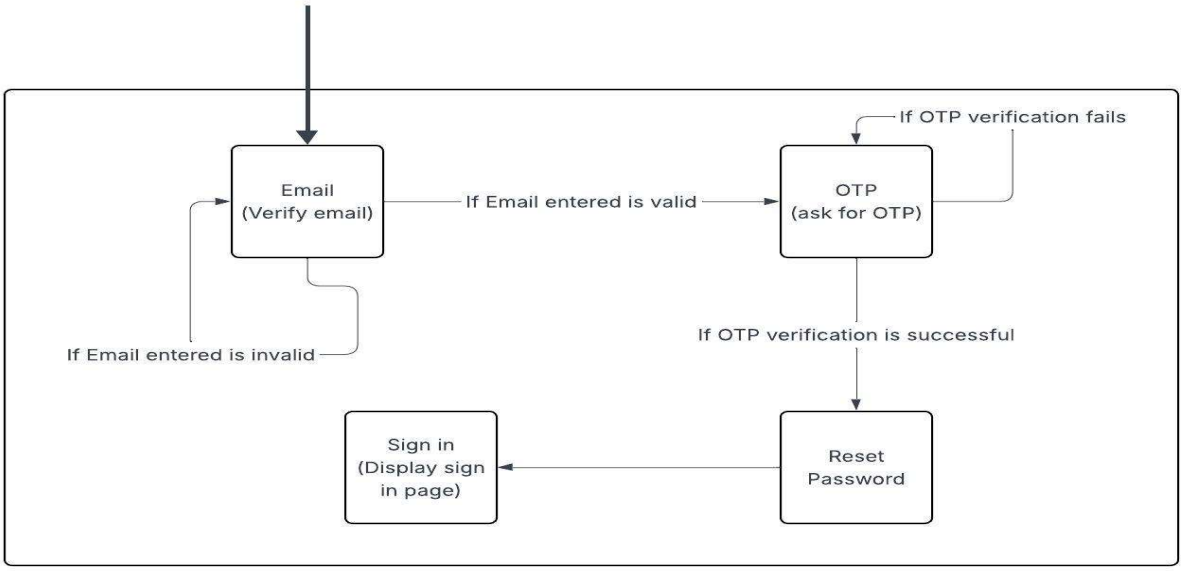
3.4.1 Overall State Diagrams



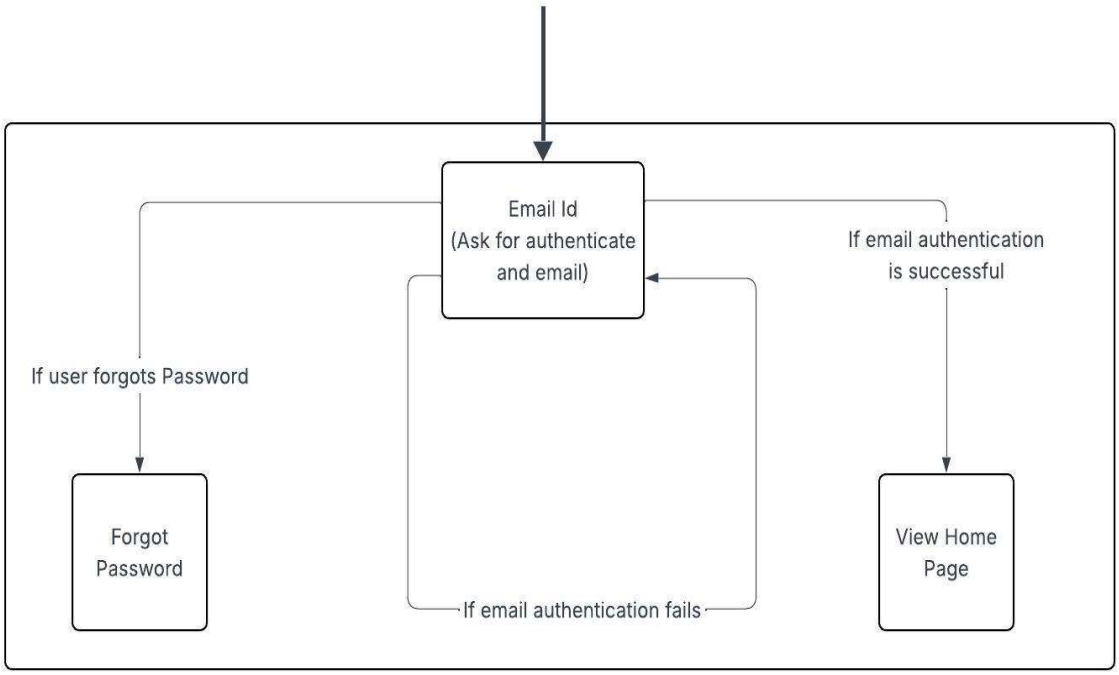
3.4.2 SIGN UP (FOR A NEW USER)



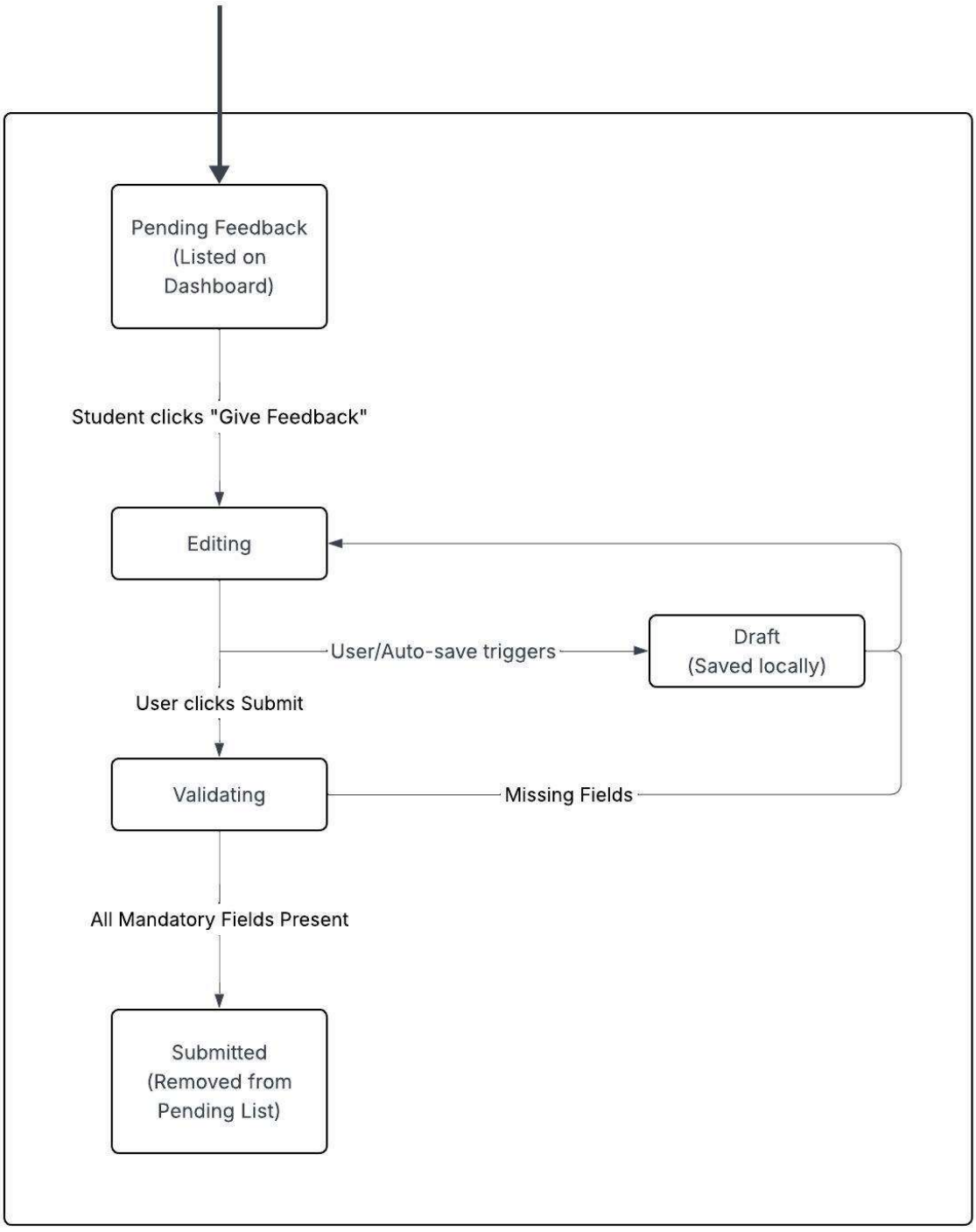
3.4.3 FORGOT PASSWORD



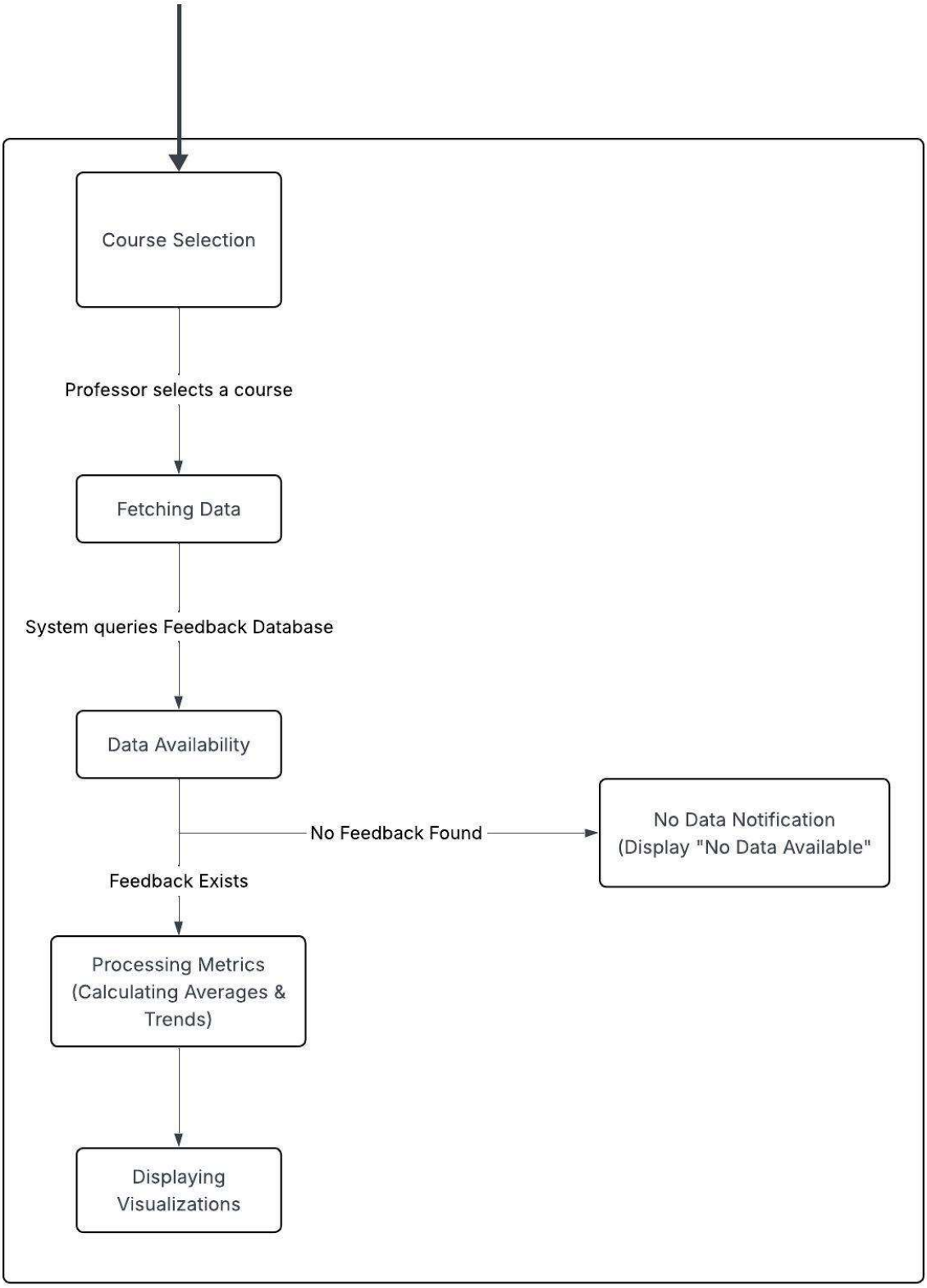
3.4.4 SIGN IN


Email Id
(Ask for authenticate
and email)
If email authentication
is successful
If user forgots Password
Forgot
Password
View Home
Page
If email authentication fails

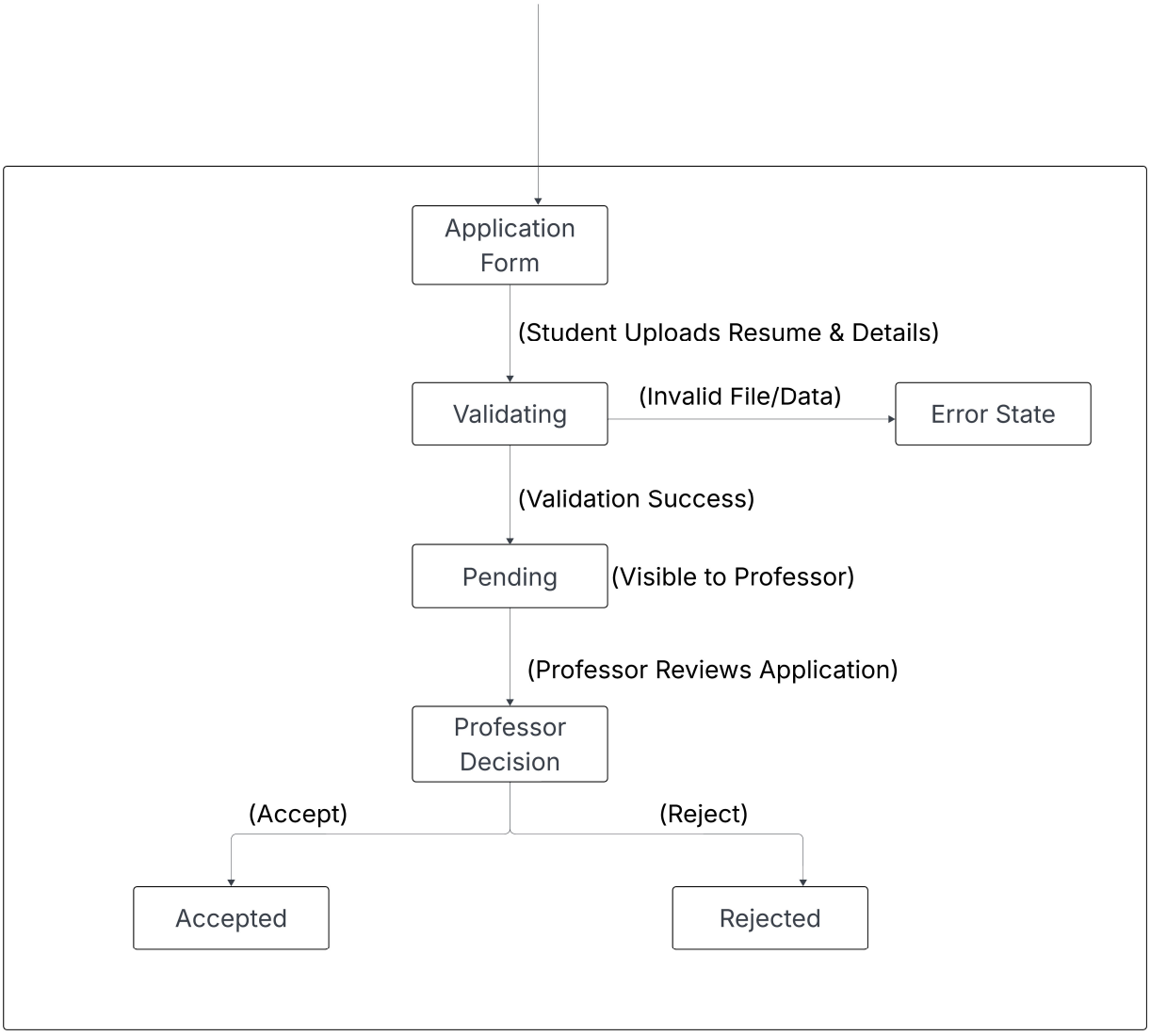
3.4.5 Student Lecture Feedback Submission:



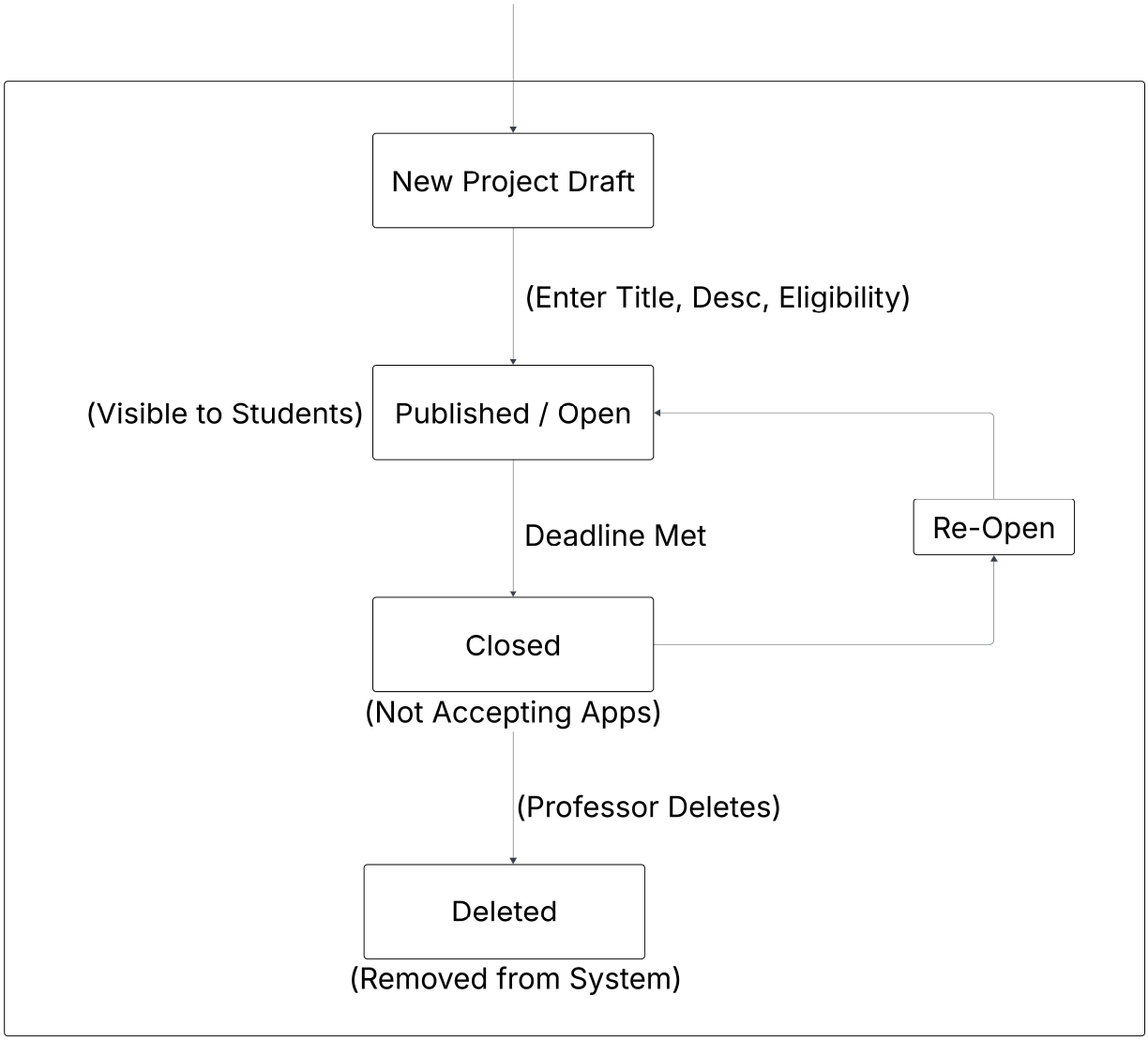
3.4.6 View Course Analytics



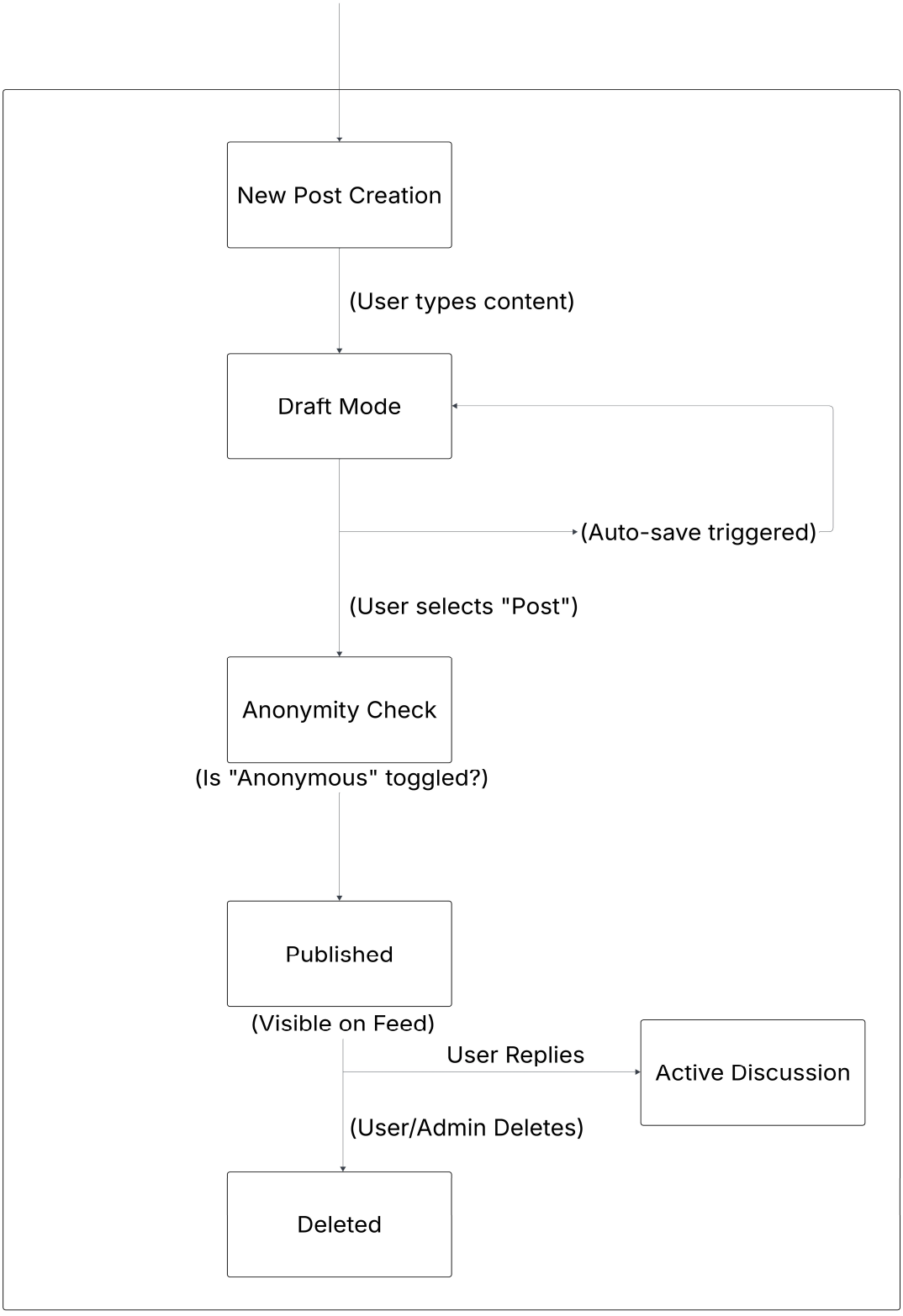
3.4.7 Student Project Module



3.4.8 Professor Project Module



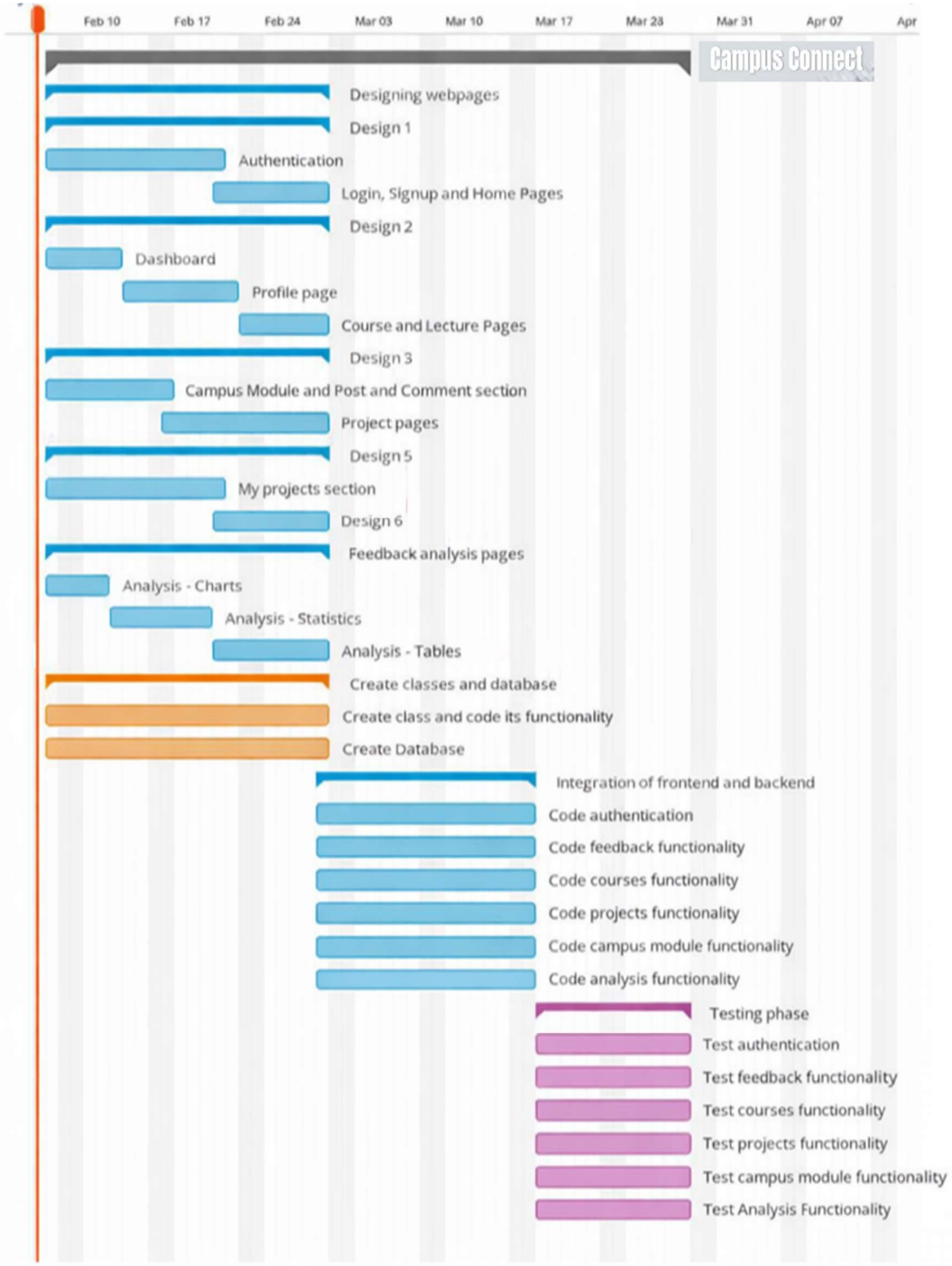
3.4.9 Campus Interaction Module



4 Project Plan

4.1 Project planning

4



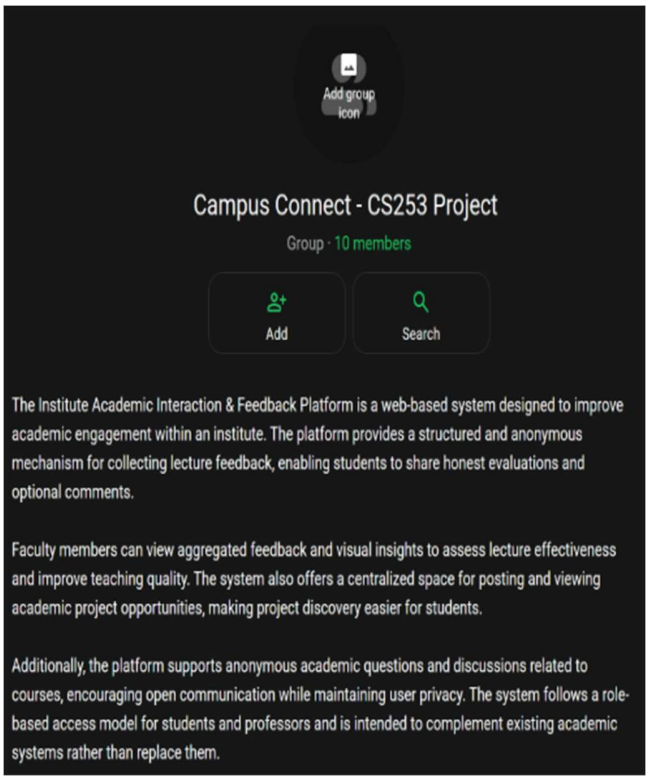
Task	**Assigned To**
Design 1 (Login, Signup & Home Pages)	Navdeep Singh
Design 2 (Student Dashboard & Profile Page)	Raj, Dishank Kumawat
Design 3 (Courses & Lecture Pages)	Mrityunjay Kumar, Rupam Das
Design 4 (Campus Feed & Post/Comment UI)	Prateek Raj, Karan Kumar
Design 5 (Project Pages UI)	Sumit Vishwakarma
Design 6 (My Projects Section & Feedback UI)	Vatsal Kumar, Prabhakar Raj
Create Classes and Database	Vatsal Kumar
Backend – Authentication	Navdeep Singh, Dishank Kumawat
Backend – Courses & Content Module	Karan Kumar
Backend – Campus Feed (Post & Comment System)	Raj, Rupam Das
Backend – Feedback System & History	Mrityunjay Kumar, Sumit Vishwakarma
Backend – Project Module	Prateek Raj
Backend – Analysis & Reports	Vatsal Kumar, Prabhakar Raj

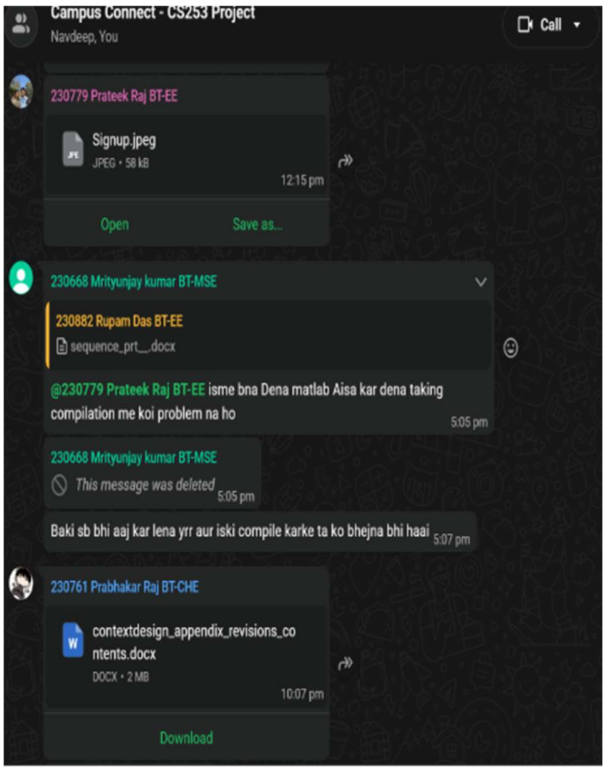
4.2 Code Collaboration

We are using GitHub to collaborate on the code. Separate repositories have been made for the frontend User Interface codes and backend (main) repository to store the database and applications.

4.3 Communication

We primarily used a WhatsApp group for seamless coordination and held offline meetings twice weekly to ensure steady progress. These meetings focused on key updates, solutions, and aligning efforts toward the next goals. At the end of each session, we outlined and assigned the tasks for the next meeting, ensuring clarity, accountability, and a well-structured workflow.


Add group icon
Campus Connect - CS253 Project
Group · 10 members
Add
Search
The Institute Academic Interaction & Feedback Platform is a web-based system designed to improve academic engagement within an institute. The platform provides a structured and anonymous mechanism for collecting lecture feedback, enabling students to share honest evaluations and optional comments.
Faculty members can view aggregated feedback and visual insights to assess lecture effectiveness and improve teaching quality. The system also offers a centralized space for posting and viewing academic project opportunities, making project discovery easier for students.
Additionally, the platform supports anonymous academic questions and discussions related to courses, encouraging open communication while maintaining user privacy. The system follows a role-based access model for students and professors and is intended to complement existing academic systems rather than replace them.


Campus Connect - CS253 Project
Navdeep, You
Call
230779 Prateek Raj BT-EE
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12:15 pm
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230668 Mrityunjay kumar BT-MSE
230882 Rupam Das BT-EE
sequence_prt__.docx
@230779 Prateek Raj BT-EE isme bna Dena matlab Aisa kar dena taking compilation me koi problem na ho
5:05 pm
230668 Mrityunjay kumar BT-MSE
This message was deleted
5:05 pm
Baki sb bhi aaj kar lena yrr aur iski compile karke ta ko bhejna bhi haai
5:07 pm
230761 Prabhakar Raj BT-CHE
contextdesign_appendix_revisions_contents.docx
DOCX • 2 MB
10:07 pm
Download

Appendix A - Group Log

S.No	Date	Timings	Venue	Description
1.	25/01/2026	18:00 to 19:00	Google Meet	Discussed SDS format and finalized section division among team members
2.	26/01/2026	17:00 to 19:00	Hall 12	Prepared outline for Context Design and started Context Model drafting.
3.	28/01/2026	19:00 to 23:00	Hall 12	Progressed with Human Interface Design and discussed UI layout flow.
4	30/01/2026	18:00 to 19:30	Hall 12	Worked on Architecture Design section and identified major system components.
5.	02/01/2026	18:00 to 20:00	Hall 12	Worked on Object-Oriented Design section and started Use Case Diagram preparation
6.	04/01/2026	18:00 to 21:30	Hall 12	Worked on Class Diagram and identified system classes and relationships.
7.	05/01/2026	18:30 to 23:30	Hall 12	Prepared Sequence Diagram and State Diagram flow for key modules.
8.	06/01/2026	13:00 to 15:00	Hall 12	Final review meeting to compile and organize all SDS sections, merge diagrams and content, and complete remaining formatting work.